

Universe24: Physical Relations from the Group Ring of a Cyclic Group under the Octahedral Group of Order 24

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Abstract

The paper has one algebraic object: the integer group ring of a cyclic group, the phase discrete, N_ϕ steps, on a cubic lattice whose symmetries are the 48 signed permutations, their rotations the group of order 24; six bounded integer operations on it act at every Node and interval. Outside are detectors and clicks only; a record ends at one comparison, the click, the one non-local step. An amplitude is an integer sum of N_ϕ -th roots of unity, Born's form its square at the click under one axiom of the apparatus, Born's rule its limit; interference counts, marginals and CHSH are exact rationals. Exact on the lattice: the books, Gauss's law, every direction's pace, the CHSH sum 181/64 from 512 to 8192, 1.05 standard errors from the measurement. Recovered in limits: $c = 1/\sqrt{3}$, Tsirelson's value, and under a shell average Newton's inverse square and Poisson's equation. Under two named hypotheses on the click, Lorentz's factors and Einstein's step are reached, the equivalence principle under the first alone, not derived from nothing; the law as built meets the first only. Everything compared with nature is a count between clicks at a detector or a ratio of such counts; nothing inside the board is compared; of ten observables predicted, four agree, six disagree; eight are not predicted by construction. The bending's $2(1 + \gamma)$ and the atom's $1/j^2$ ladder are conjectures. The law as built has no relativistic dynamics; its sequential wheel signals in a pair's order, not its counts.

1 Introduction

The algebra first. This paper begins with one algebraic object and states what follows from it. The object is the integer group ring of a cyclic group, the phase grain N_ϕ , carried on a cubic lattice whose symmetries are the 48 signed permutations of the axes, their rotations the group of order 24 (Theorem 1); the law is six integer operations on that ring, and every result the paper calls exact is an identity of it under declared tables and world conditions (Theorems 4 to 8, the conservation books, Gauss's law, the pace of every direction). Every quantity compared with nature is a count between clicks at a detector or a ratio of such counts; a reading of the board itself is a diagnostic and is compared with nothing (the reading rule below; Section 7.1; Table S1). From the same object, with the hypotheses on how a click reads it, one chain reaches above the board: under (A1) with (A3), locality Outside, the r -free lines of Lorentz's factors, and under (A2) their scale and Einstein's step; under (A1) and five assumptions of the law as built, with no (A2), the equivalence principle and, in a limit under a shell average, Newton's inverse square (the paragraphs 'Newton from the small step' and 'The road to each'): reached from the algebra, not derived from nothing, and the law as built meets (A1) and not (A2), so what the runtime shows of that chain is read row by row in Table S1.

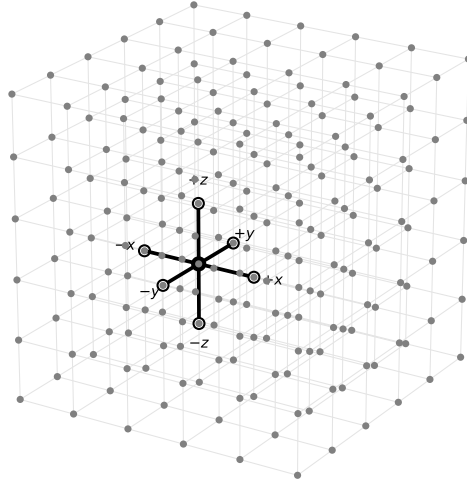
The paper in one page. The law is one map of bounded integers at every Node and interval, Eq. (1), its rates and walls made of the six operations; Outside there are detectors and their clicks only, and the GameBoard is read in no other way (P11). What is exact on the board, what is recovered in a named limit and what no rule forces are listed once, in ‘What is proved’ (Section 4). Against nature, of the ten observables the law as built predicts (eleven once row 14 is read), four agree within tolerance (1a, 2b, 9; 5a as a bound met), six disagree (1c, 6, 7b, 8a, 8b, 13), and eight are not predicted by construction (3, 4a, 4b, 5b, 7a, 8c, 11a, 11c), each with its missing piece named in its cell and its kind, the instrument missing in the model, nature’s number a bound or a fitted parameter, no click definition yet, or the model’s own number a declared input; 2a NOT COMPARED until its source is verified; six rows keep their words (12 history, 2c not compared, 2a as said, 10 not yet, 11b a pin, 14 not yet read against nature), and 1b is a control, not counted; the hypotheses beside rows 6, 13 and 14 change no verdict of the law (Table S1), the failures the law’s own. Two relations enter as conjectures from the algebra, labelled so and distinguished from what a detector reads: the bending’s coefficient $2(1+\gamma)$ on the one constant, and the atom’s ladder $1/j^2$ with Balmer’s $27/20$; the ring under its rule is read and its constant does not close within the grain, and the ladder is not read on the board, no lattice able to hold the atom’s scale being run. The law as built meets (A1) and not (A2), so it recovers no relativistic dynamics; and its sequential wheel signals in the order of the pair’s outcomes, though not in their counts. One finite-grain value, $S = 181/64$, inside Poh et al. at 1.05 standard errors, a prediction once the wheel meets no-signalling in the order of its outcomes. The method is one platform, by the algebra and by the simulator alike: an Inside step on a family’s attributes, the conversion by an emitter and a detector, an Outside form as a ratio of counts, and a detector’s reading pinned before the run.

The general formula. The law is one map Φ applied at every Node of the GameBoard at every interval, per component s of the state, each with its own rate r and wall d :

$$\boxed{s \leftarrow s + r, \quad e \leftarrow \text{sign}(s) \min(\lfloor |s|/d \rfloor, a), \quad s \leftarrow s - e d.} \quad (1)$$

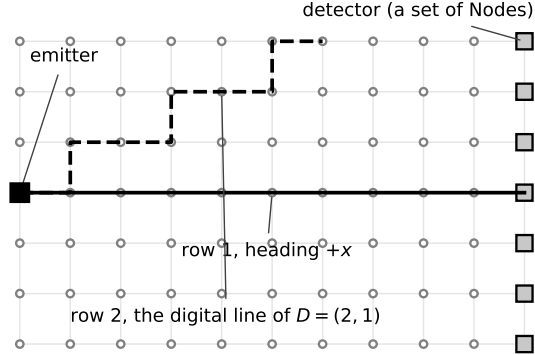
Each component of the state is a bounded integer accumulator with a declared rate and a declared wall: it is translated by its rate, counted in walls (capped at $a = 1$ for the drive, uncapped elsewhere), and reduced by the walls it holds; every wall counted is an event, a Link crossed, a phase step taken, a count completed, a birth, a push, and nothing else happens. The rates and walls are made of six integer operations and nothing else. They are the translation; the multiplication by a declared integer matrix or bilinear form; the sum in the group ring of the phase circle; a permutation; the evaluation of a record at the roots of unity with its norm; and the division with the remainder kept, together with the comparison that is the event. One comparison, the click, is the only read-out. Only a detector’s reading is a measurement. Section 5.1 gives the full definition; every derivation of this paper names which of these rules it starts from.

Why physics behaves like modern algebra. The claim of the paper is that the forms of several different physical phenomena are recovered from this one discrete update law and a small number of assumptions, Inside by derivation and Outside at the clicks, and the reason is stated in three steps that the paper then follows. First, Outside is arithmetic in $\mathbb{Q}$: every measured quantity is a count at a click or a ratio of two counts (a velocity is Nodes apart over counts apart; a visibility, a correlation and a redshift are ratios of counts), and nothing measured is a real number. Second, Inside is integer and local: six operations on bounded integers at a Node and its six neighbours, each rule generic, vector and local. Written down, the rules put into the cells are the cyclic group of the phase, the group ring of the arrivals, the integer matrices of the split and the rotation, the



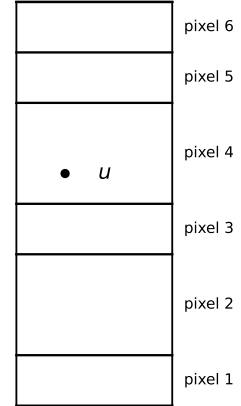
one Node and its six Ports, the Links to its six neighbours

Figure 1: The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Figure 3. Nothing sits at a Node beyond the rows there. Drawn from the definitions; no run.



Inside: one Link per interval, the phase turned per Link

the rungs b_k of the ladder



Outside: the click

Figure 2: The mechanism, drawn from the definitions and no run. Inside (left): a GameBoard of Nodes joined by Links; an emitter puts one record of two rows on the board, each row on the digital line of its direction at one Link per interval, its phase turned per Link, nothing kept at a Node; a detector is a set of Nodes. Outside (right): the click, one comparison of the record's birth phase u against the rungs of its ladder (Eq. (4)); the cell u falls in is the outcome, and the record is deleted. Nothing else is read.

evaluation at the roots of unity and the translation group's shift. So the state is a free $\mathbb{Z}$ -module and the interval a $\mathbb{Z}$ -linear map on it (Section 5.1). Its couplings are tables: the circle's C and S , the split's and the rotation's integer matrices, the family table. Born's rule is a coupling computed

from N_ϕ , the rungs of Eq. (4), so a click is a comparison of integers. Third, the transformations between click families that preserve the passage of information are the Lorentz group up to scale, the click theorem's [48] (Section 3), so the symmetry Outside has is the algebra's and not a metric put in. Every result the paper calls exact is an identity of that algebra under its declared tables, matrices and family table, which are inputs (Table S5). Each is checked against the register's readings by kind and not chosen to fit a reading; where a form was declared, the paper says so. The decisive question is how much of that the law really forces: if every phenomenon needed a rule that puts its result in beforehand, the law would explain nothing; if the same rules force several results with little freedom, that is the contribution. So the paper keeps one ledger (Table S5): for every result, the rules of Eq. (1) it starts from, the assumptions added with their kind, the freedom left and the ground of the derivation. What can be established now is an explicit model with exact and conditional results and checks after a detector. The paper is built around the distance between that and the broad claim that Outside represents nature.

Where the object came from. The object was not chosen first. One Node with six Ports, one Link per interval and bounded integers were the assumptions of a simulator; its rules, written one by one as operations on a state vector, turned out to be six maps on one ring, and the maps that preserve a Node and its causal cone turned out to be the 48 signed permutations, their rotations the group of order 24 (Theorem 1); what did not converge to that group and that ring was not put in. The road, from the postulates to the objects they force and back, with the simulator running first and the derivations following it, is Section 6; the GameBoard the object is computed with is Section 5.

The reading rule and the notation. Only a detector's click is a measurement, and a click is an action of the law on the state, the one step that deletes Inside and the one row that Outside gains (the frame's assumption, P11; the law's rule [20, section 10]): a click, a record's moments or an external thing's reading, as the world file declares. Every physics name in this paper (Gauss's, Newton's, Coulomb's, Born's, Planck's, Tsirelson's) names the known form that the law's result recovers, Inside by derivation or Outside at a detector. A number of the GameBoard's state is a diagnostic of the host, never compared with nature and never pinned. Every symbol is named in English at its first use, a scalar plain, a vector in bold lowercase, a matrix or an operator in bold uppercase; the symbol table is Table S3 of the Supplementary Material, the program's glossary is on the tree [15], and every number has its one source [5].

What this paper contributes, and what it does not claim. New here: one explicitly stated update rule with its implementation, run from world files with every reading pinned before the run; exact conditional algebraic results on the lattice (the quadratic read-out characterization, the count marginals, the CHSH sum as an exact rational of the grain, 181/64 on the registered plateau); one platform on which every formula is reached by the algebra and by the simulator alike; and a confrontation register that states the failures with the same care as the passes. Not claimed: a derivation of the Lorentz group from the six operations, the atom's levels as a reading, a unique prediction of 181/64 without a rule that fixes the grain, or any comparison with nature beyond the rows of Table S1, each read by kind.

2 The algebra: the group, the ring and the families

This section states the object and nothing else: the words of the board, the group of the six Ports, the ring of the phase and the record as its element, the rules of one interval, the click that reads a record once, and the families as declared integers on those objects. The definitions are gathered here from the sections that used them; every later section cites them by number.

The definitions, in order. Five objects, each one line of mathematics before its words, and the names used from here on.

- (1) The GameBoard: the Nodes $\mathbb{Z}^3$, a box whose axes are periodic or open; the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$, with the involution $e \mapsto -e$ (the opposite Port); one Link per interval.
- (2) The group of order 24: $G_{48} = \{g : P \rightarrow P \text{ a bijection with } g(-e) = -g(e)\}$, the 48 signed permutations of the three axes, the symmetry group of the cube; $\det : G_{48} \rightarrow \{+1, -1\}$ a homomorphism; $G_{24} = \ker \det$, the rotations, of index 2: the group of order 24, the name used throughout (Theorem 1 states the symmetric group it is isomorphic to). The hand, a pseudoscalar, $h \mapsto \det(g) h$, tells the two cosets apart.
- (3) The phase circle and its ring: $\mathbb{Z}_{N_\phi} = \mathbb{Z}/N_\phi\mathbb{Z}$, the cyclic group of the phase, N_ϕ declared per world; $\mathbb{Z}[\mathbb{Z}_{N_\phi}] = \{\sum_p f_p x^p : f_p \in \mathbb{Z}\}$, its integer group ring, the addition by components and $x^p x^q = x^{p+q}$; a record's rows at one Node and label are one element f of it (the paragraph "The objects" and Definition 1).
- (4) The six operations, maps on those objects: (T) $s \mapsto s+r$; (B) $(\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u}^T \mathbf{C} \mathbf{v}$ with $\mathbf{C}$ declared; (G) $(f, g) \mapsto f + g$ in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$; (P) a permutation of the joint state; (E) $\text{ev} : \mathbb{Z}[\mathbb{Z}_{N_\phi}] \rightarrow \mathbb{Z}[\zeta_{N_\phi}]$, ζ_{N_ϕ} the primitive N_ϕ -th root of unity and $\mathbb{Z}[\zeta_{N_\phi}]$ the cyclotomic integers; (D) $s \mapsto (e, s - ed)$ with the comparison. The rules of one interval are made of them (Definition 2); each is written out below ("The six operations as maps, in symbols").
- (5) The click: $\text{ev}(f)$ with its norm $|\text{ev}(f)|^2$ at a detector, the one read-out (Definition 3); and Outside, the counts between clicks and their ratios, the only things compared with nature (Section 3).

The postulates, by name. The eleven postulates of the law keep the numbers under which Section 5.1 states them and get a name here, used beside the number from now on: P1 bounded integers; P2 the cubic lattice, six Ports; P3 the six operations; P4 locality, a rule reads its own record and the six neighbours and nothing is kept at a Node; P5 one Link per interval; P6 one read-out, a record read once by one comparison; P7 the declared tables; P8 the families as integers; P9 the blind flight, a row in transit moves at its declared rate whatever its phase and whatever the crowd, no dispersion; P10 the click axiom, the click's weight the one positive quadratic form; P11 clicks only, the GameBoard read in no other way. What each forces in the algebra, and what each object gives back, are the two tables of Section 6.

The words of this paper. The *GameBoard* is the cubic array of Nodes, each joined to six neighbours by Links through directional Ports; time is a sequence of intervals. *Inside* is the board and what moves on it, where no one measures; *Outside* is the game above the board, the emitters, the detectors and their clicks, the only place a number is read. A *row* is one message on the board

(a Node, a direction, an age, a phase, an amount, a multiplicity and its record); a *record* is the rows born together at one emitter, one quantum. The *click* is a detector's one comparison, which reads a record once and deletes it: the only read-out, and the one non-local step. A *rung* is one integer threshold of the click's ladder over the birth phase (Eq. (4)); the cell the phase falls in is the outcome. Every number carries its *kind*: DETECTOR, a count of clicks, the only measurement; GAMEBOARD, a number of the board's state, a diagnostic and never a result. Figure 1 draws the board in space, Figure 2 the two sides.

The six Ports, the group and the front. Everything geometric in the law follows from one Node and its six Ports, $\pm x$, $\pm y$, $\pm z$, and from one Link per interval. The maps of the six Ports onto themselves that keep opposite Ports opposite are the signed permutations of the three axes: $3!$ orderings of the axes times 2^3 choices of sign, 48 in all, the symmetry group of the cube and of the octahedron (Theorem 1). The determinant of a signed permutation is ± 1 , so it splits the 48 into two halves of 24: the rotations, of determinant $+1$, which are the permutations of the cube's four body diagonals; and the improper maps, of determinant -1 , the reflections with the central inversion among them. The hand of a row is that determinant made a column of the state: one pseudoscalar bit that a map g sends to $\det(g)$ times itself, so a world with a hand reads which half a map belongs to. The same six Ports bound the pace (Proposition 1 below); that the rows' pace c sits at that bound and not below is the flight's declared wall T_D , a rule chosen among few (P9), and given the wall the value is the flight operator's norm, attained exactly on the body diagonals (the derivation's 24.1, row 9).

Theorem 1 (the 24 of the name). *The maps of the six Ports onto themselves that send opposite Ports to opposite Ports form the group of the signed permutations of the three axes, of order $2^3 \cdot 3! = 48$. The determinant is a homomorphism of this group onto $\{+1, -1\}$; its kernel, the group of order 24 (the rotation group of the cube), has index 2 and is the group of the permutations of the cube's four body diagonals, the symmetric group S_4 ; the other coset, the reflections, has 24 elements too. The hand, the one pseudoscalar column of a row, which a map g sends to $\det(g)$ times itself, tells the 24 rotations from the 24 reflections, as does an axial vector.*

The proof is the Supplementary Material, S5.

The objects. A record's rows at one Node and label are one element $f = \sum_p f_p x^p$ of the group ring $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, x the generator of one phase step and f_p the amount at the phase p ; the pointer is $\text{ev}(f) = \sum_p f_p \zeta^p$, ζ the primitive N_ϕ -th root of unity, at the tables' scale. The merge's cancel identifies $x^{N_\phi/2}$ with -1 , so the rows after the cancel are an element of $\mathbb{Z}[x]/(x^{N_\phi/2} + 1)$; for N_ϕ a power of two (every registered N_ϕ) that quotient is the ring of cyclotomic integers $\mathbb{Z}[\zeta_{N_\phi}]$ itself, a free $\mathbb{Z}$ -module of rank $N_\phi/2$, and ev is the identity on it: the GameBoard's cancel and the ideal click's zero, the evaluation on $\mathbb{Z}[\zeta_{N_\phi}]$, are one relation, exactly (for an N_ϕ with an odd factor they are not; the register has no such N_ϕ). The table click, Definition 3 with the declared integer tables, is a different map: the built click keeps the relation one way: every cancel reads $(0, 0)$, and the tables read $(0, 0)$ on more, the table pointer being a $\mathbb{Z}$ -linear map of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ onto a lattice of rank 2, its kernel of rank $N_\phi - 2$ against the cancel's $N_\phi/2$; at $N_\phi = 8192$ only 1952 of the 8192 table pairs are distinct and two rows of amount 1 at the phases 0 and 4097 read $(0, 0)$ where $|\text{ev}(f)|^2 = 4 \sin^2(\pi/8192) = 5.9 \times 10^{-7}$; at $N_\phi = 64$ no zero exists with at most three rows of amounts up to 4 ([34], a computation from the tables, not a run), a fact of the tables and not a relation of the module. The reading is a function $R : \mathbb{Z}[\zeta_{N_\phi}] \rightarrow \mathbb{R}$ of a cell's element, up to one factor per record, which the multiplicity $\mathfrak{m}$ books.

Definition 1 (Row and record). A *row* is $(x, D, \tau, p, e, h, r, \ell, w, \mathbf{m})$: Node, direction, age, phase $p \in \mathbb{Z}_{N_\phi}$, emitter, content per unit, record r , branch label ℓ , amount $w \geq 1$, multiplicity $\mathbf{m} \geq 1$. Its amplitude is

$$z = \frac{w}{\sqrt{\mathbf{m}}} \frac{C_{N_\phi}[p] + iS_{N_\phi}[p]}{256}, \quad |z|^2 = \frac{w^2}{\mathbf{m}} \cdot \frac{C_{N_\phi}[p]^2 + S_{N_\phi}[p]^2}{65536}. \quad (2)$$

Two rows equal in every field but w are one row with the amounts added; two rows of one record and label equal in every field but a phase difference of $N_\phi/2$ are one row with the amounts subtracted (an equal pair is no row). The state of record r is an element ψ_r of the free $\mathbb{Z}$ -module M_r on the basis $(x, D, \tau, p \bmod N_\phi/2, e, h, \ell, \mathbf{m})$ with $|p + N_\phi/2\rangle = -|p\rangle$; the scaling $(w, \mathbf{m}) \rightarrow (kw, k^2\mathbf{m})$ leaves the amplitude and the norm unchanged and is an identification at the read-out, not a relation of the module (the quotient by it would not be free, $b_1 = 2b_4 = 4b_{16} = \dots$ for the rows $b_{\mathbf{m}}$ of amount 1); the GameBoard's state is one normal form per element of $\bigoplus_r M_r$, and the multiplicity, which every split multiplies by A , is bounded by the law's bound of an amount, a multiplicity and a momentum component, $2^{62} - 1$ (`events/world.py`, one bit spared for one more sum), inside the host's working bound $2^{63} - 1$ for every intermediate (`core/integer.py`); a run that would pass either is refused (P1's bound, named here). In the row form of Theorem 4 a rotation multiplies $\mathbf{m}$ by 65536, so four in succession would pass the law's bound; the engine does not carry that factor, keeping a record's label set and applying the tables once at the click's form, the cell's weight reduced by its greatest common divisor, so the depth of successive rotations is not limited by the bound; where the bound binds, it is a limit of the implementation's domain and never a prediction. The norm is $\langle \psi, \psi \rangle = \sum_{\text{rows}} w^2/\mathbf{m}$. The archived engine narrows the quotient at a detector: rows of one record with different $\mathbf{m}$ at one set are added when the ratio of the multiplicities is the square of a rational (the pointers rescaled by the integer roots) and refused otherwise; at the tree of the runs two multiplicities at one set were refused, and no reported world has them, so every set below reads one $\mathbf{m}$.

The host, not any Node, keeps per live record the birth phase $u \in \mathbb{Z}_{N_\phi}$, the count of live units (amounts), an accumulated pointer (X, Y) per (Node, label) and a gathered flag. This is the apparatus's layer; the GameBoard never reads it. It is a memory no Node could own: on the two-slit world the two paths of one record reach a pixel up to 18 intervals apart, so no Node ever holds both rows [6].

Definition 2 (The update rules of one interval). Applied in this order to every row.

- F Flight: $(x, D, \tau, p) \mapsto (x + \text{pos}_d(\tau + 1) - \text{pos}_D(\tau), d, \tau + 1, p + \varphi(\tau))$, where the family's phase turn is either an integer per Link, added only when the step is a Link, or a pair $[n, d]$ per interval of age, $\varphi(\tau) = \lfloor (\tau + 1)n/d \rfloor - \lfloor \tau n/d \rfloor$, counted from the row's birth or last re-emission.
- C Collision: at a Node holding several rows of amount 1 the table permutes their directions as a function of the whole set present; $r, \ell, \mathbf{m}$ unchanged; a bijection of states, not a linear map. No collision acts in any world of this paper.
- S Split, at a measured event with outputs $\{(d_i, a_i, t_i)\}$, $a_i \in \mathbb{Z}$: the row $(w, \mathbf{m}, p)$ is absorbed and re-emitted as the rows $(wa_i, \mathbf{m}A, p + t_i)$ on d_i at age 0, $A = \sum_i a_i^2$. The balanced splitter is $(1, 1)$, $A = 2$; the reflection's turn is $N_\phi/4$.
- R Rotation, at a set with setting s and turn t , on a label pair $(0, 1)$ read into the channels $(+, -)$: $U_{s,t} = \begin{pmatrix} C'[s] & S'[s]v(t) \\ -S'[s] & C'[s]v(t) \end{pmatrix}$ with $v(t)$ the turn by t steps on the label-1 column; a row of label 0 becomes $(wC'[s], 65536\mathbf{m}, p)$ on $+$ and $(wS'[s], 65536\mathbf{m}, p + N_\phi/2)$ on $-$, a row

of label 1 becomes $(wS'[s], 65536\mathbf{m}, p+t)$ on $+$ and $(wC'[s], 65536\mathbf{m}, p+t)$ on $-$ (the row form exact for t a multiple of $N_\phi/4$, the only turns used here; the archived code multiplies the column by the tables' phasor at t). With $t = 0$ this is U_s of Theorem 4; the GHZ bases below are $X = (s, 0)$ and $Y = (s, N_\phi/4)$.

M Normal form: merge and cancel as in Definition 1.

B Birth: a lamp's self-creation at interval t releases record r with u the lamp's phase at the birth, $(b-1) \bmod N_\phi$ for a lamp started at 0 that turns one step per birth, b its count of births (every world here; $b-1 = t-1$ only while the lamp births at every interval; the archived code writes $(b-1) \bmod N_\phi$ directly), rows of amount 1 on its k directions with $\mathbf{m} = k$, each row's phase starting at u (plus the lamp's own turn), and, for a pair, the declared labels with integer weights on separate arms.

The six operations as maps, in symbols. Every rate and wall of Eq. (1) is made of six maps on bounded integers and nothing else. Each is written here as the map it is, the statements of the algebra document [10] with the letters of the derivation record [22], on the objects above: an accumulator s , an integer on a line or a circle; a record's element $f \in \mathbb{Z}[\mathbb{Z}_{N_\phi}]$; a vector of $\mathbb{Z}^3$ or $\mathbb{Z}^k$.

- (T) *The translation*, $s \mapsto s+r$ on $\mathbb{Z}^k$ or on a torus: every count of the law, the flight on the digital line, the phase's turn, the age, the drive, the owed count, the release, the lamp's wheel, the push per column. Where r is a constant of the world the closed form is $s(t) = \lfloor s_0 + rt \rfloor$. On a row the flight's accumulator gains $2S_1N_l$ per interval ($S_1 = |D_x| + |D_y| + |D_z|$ the direction's Manhattan length, N_l the label's scale, 64 on the register) against the wall $2T_D$, started at T_D , so the Links crossed by the age τ are $m_D(\tau) = \lfloor (2\tau S_1N_l + T_D)/(2T_D) \rfloor$ and the k -th Link falls at $\tau_k = \lceil (2k-1)T_D/(2S_1N_l) \rceil$, the flight table's closed form; on the phase a row turns $\varphi(\tau)$ steps per interval of age (Definition 2, F), read modulo N_ϕ . The translation group of the box, $\mathbb{Z}_X \times \mathbb{Z}_Y \times \mathbb{Z}_Z$, is this map on Nodes.
- (B) *The bilinear form with a declared matrix*, $(\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u}^\top \mathbf{C} \mathbf{v}$, $\mathbf{C}$ an integer matrix declared at load: the push, whose rate is the reader's content times the label flow $\mathbf{a}$ of the arriving rows through the coupling matrix, the columns with their signs (Section 4.2); the split and the rotation, integer matrices on the rows (Definition 2, S and R); and the click's weight, $\mathbf{f}^\top \mathbf{G} \mathbf{f}$, $\mathbf{f}$ the coordinate vector of f on the basis x^p and $\mathbf{G} = \mathbf{E}^\top \mathbf{E}$ the tables' Gram matrix, nothing squared as a step of its own.
- (G) *The group-ring addition*, $(f, g) \mapsto f + g$ in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$: the merge of a record's rows at one Node, with the cancel $x^{p+N_\phi/2} = -x^p$, the passage to the quotient $\mathbb{Z}[x]/(x^{N_\phi/2} + 1)$, which for N_ϕ a power of two is $\mathbb{Z}[\zeta_{N_\phi}]$ (the paragraph "The objects"). Two arms of a pair multiply by the ring's product, $x^p x^q = x^{p+q}$, and the labels add (Section 4.5).
- (P) *The permutation of the joint state*: a collision permutes the directions of the single units at a Node as a function of the set present (Definition 2, C; no world of this paper runs one); a gate permutes the joint labels from the control's bit; the apportioning's tie is a rotation of an order. A permutation of directions, labels or Nodes, never of contents; a bijection of states, not a linear map.
- (E) *The evaluation at the roots of unity*, $\text{ev} : \mathbb{Z}[\mathbb{Z}_{N_\phi}] \rightarrow \mathbb{Z}[\zeta_{N_\phi}]$, $\text{ev}(f) = \sum_p f_p \zeta_{N_\phi}^p$: a surjective ring homomorphism whose kernel is the ideal $(x^{N_\phi/2} + 1)$, the cancel of (G), so that $\mathbb{Z}[\mathbb{Z}_{N_\phi}]/\ker \text{ev} = \mathbb{Z}[\zeta_{N_\phi}]$ exactly; and a $*$ -homomorphism for the involution $f^*(x) = f(x^{-1})$,

the reflection of the phase, carrying f^*f , the record's autocorrelation, to $|\text{ev}(f)|^2$. The built evaluation is the tables at the scale 256: the pointer $(X, Y) = \mathbf{E}f$, $\mathbf{E}$ the $2 \times N_\phi$ integer matrix of C_{N_ϕ} and S_{N_ϕ} , a $\mathbb{Z}$ -linear map into $\mathbb{Z}^2$ that is not a ring homomorphism ($\mathbf{E}(x)^2 = (64400, 12750)$ against $256 \mathbf{E}(x^2) = (64256, 12800)$ at $N_\phi = 64$; COMPUTATION), whose kernel contains the cancel and more; its norm $X^2 + Y^2 = \mathbf{f}^\top \mathbf{G} \mathbf{f}$ is (B). The exact map is the algebra; the tables are the engine's declared rounding (P7).

- (D) *The division with the remainder kept, and the comparison, $s \mapsto (e, s - ed)$, e the event count of Eq. (1): the count is the whole part in units of the wall, signed and truncating, the remainder in $(-d, d)$ with the accumulator's sign, the Euclidean division on a non-negative accumulator (every unsigned count); every wall crossed is the event. The comparison is part of this map: the ladder's cell of Definition 3, an age against a key, a phase against a window, a threshold, each a division whose quotient is 0 or 1 and whose remainder is not read.*

One interval of a record applies, in this order, (T) on every accumulator, (D) whose whole part is the event, (B) at the push and at the click, (G) and (E) where rows are summed at one Node and where they end, and (P) at a collision: Eq. (1) is their composition, and nothing that is not one of the six enters it. Not one of the six, and not admitted: a root of the state at run time (the two run-time roots of the algebra document's chapter 2.7, the meeting's norm under the key `meeting` and the pushed row's pair under the key `optical`, later rules in no world of this paper; Lorentz's factor on a body's own counter), a rounded constant multiplying a row in flight beyond the tables declared at load, a float, a true division, a draw.

Definition 3 (The click). At a set that reads the sum, the rows of record r and label ℓ that arrive are absorbed and the layer accumulates

$$X += \sum 32 w C_{N_\phi}[p], \quad Y += \sum 32 w S_{N_\phi}[p], \quad (3)$$

the phase p as the row carries it on arrival (the turns of rule F already in it). The pointer is kept per Node: interference is coherent within one Node only, and the offer of a set of Nodes (a detector, a face) and label ℓ is the sum over its Nodes of $(X_x^2 + Y_x^2)/\mathfrak{m}$, one cell of the ladder, the click landing at the Node within the set that u selects by the same rungs; all rows at a set share $\mathfrak{m}$. When the record's live count reaches 0, with $C_K = \sum R$ over its offers and the cumulative sums $C_1 \leq \dots \leq C_K$ (the total) in the declared order,

$$b_k = \left\lfloor \frac{2N_\phi C_k + C_K}{2C_K} \right\rfloor \quad (b_0 = 0, \quad b_K = N_\phi), \quad \text{the click is the } k \text{ with } b_{k-1} \leq u < b_k, \quad (4)$$

and the world's list gains one row: the record, u , the chosen set and label, R , C_K , the count of combinations before and after. For a record with arms at two rotated sets A and B (settings a, b) the cells are the joint outcomes (o_A, o_B) in the arms' declared order, with the weights

$$R(o_A, o_B) = J(o_A, o_B)^2, \quad J(o_A, o_B) = \sum_\ell c_\ell U_a[o_A][\ell] U_b[o_B][\ell], \quad (5)$$

c_ℓ the labels' declared weights, $(1, 1)$ in every world here; and the record is gathered once, when the rows at both arms have ended. The click of an arm has two parts. The arrival: an event at the arm's own detector at its own interval, the rows absorbed and their content placed there, the flight blind to the other arm's setting and length (P9), so the arrival interval at either arm is that arm's flight alone. The outcome: the cell of u ; for the party declared first (the arms' declared order, not

the order of arrival) the coarse rung at $N_\phi/2$, a function of u alone, u the record's wheel reading fixed at its birth ($u = \text{ordinal} \times r \bmod W$), so determined before either arrival and known at the first party's arrival from the record's own wheel under the hypotheses of Theorem 7, computed by nothing at that arm and written only at the completion; for the second the fine rung within it, fixed at the completion from the joint weights. The world's row carries both outcomes at the completion interval, so the near party's outcome is written at the far party's tick, a bookkeeping of the host and not a delay of any arrival. Reading the ladder in that order is the same as a coarse rung over o_A followed by a fine rung over o_B within it with the same u , which is how Theorem 7 states it. A set that reads without absorbing (the which-path read of the polariser) stores the record's residual selector keyed by its labels and lets the rows go on; it is a deferred offer, gathered at the record's one click on the labels present then, not an outcome at the interval of the read. A read, then a rotation of the labels, then a second read, taken as two outcomes in time, is not modelled: the first read's selector is applied once, at the gather, to the labels as the rotation left them [19].

Lemma 1 (One click per record). *A record every row of which ends (at a set, a face or the border; not a row parked at rest or on a periodic axis without a lifetime) completes and is gathered exactly once: the live count reaches 0; $b_K = N_\phi$ (with $C_K > 0$, the born row's norm up to the tables' rounding, Theorem 4; at $C_K = 0$ the engine gathers nowhere, [6, record 363]) and $b_0 = 0$, so every $u \in \mathbb{Z}_{N_\phi}$ lies in exactly one cell. Every record of the worlds below is of this kind; the delay between a record's last arrival and its completion is the record's, not the GameBoard's.*

Three things are put in here and not derived, and the paper claims none of them. Definition 3 recovers the Born rule's form: the weight is the squared length of the summed pointer, and the ruler over u is inverse-transform sampling of a uniform variable, so over births with u uniform on $\mathbb{Z}_{N_\phi}$ the count in channel k is $b_k - b_{k-1}$, the difference of two cumulative rungs, Born to the resolution $1/N_\phi$ per cell. No principle below the square is offered. The uniformity of u is an assumption about the source: $u = (b - 1) \bmod N_\phi$ is uniform over N_ϕ consecutive births, (one birth per u in every run below), and its independence of the settings is a fact of a world, not of the model: u is a clock and a chooser is a clock, so they are independent when the chooser's period is coprime to N_ϕ and the run covers their common period, as in this paper's choosers' world (periods 3 and 5 against $N_\phi = 64$). The tables at 1/256 are the third of these three, and a declared input of the law (Section 5.1, P7). What the paper derives is what follows from these with the GameBoard's rules: the interference of one record, the exact marginals, the $\cos^2$ correlation of the pair, GHZ's zeros, and the finite- N_ϕ value of S . The click is the only step that removes combinations and adds a row to the world's list; the birth is the only step that adds combinations.

The families as declared integers. Every family of the register is one row of declared integers (P8) and nothing else: the quantum h (1 on a free family, whose rows carry their energy as turns, 0 on a paid family, whose content is the mass), the charge per unit of content as a rational pair, the columns with their signs, the lifetime, whether the phase turns, the hand; the content M of a body is the world's declaration. Written down, a family is where its quantities live: the content on $\mathbb{Z}$, the non-compact scale, free; the phase on $\mathbb{Z}_{N_\phi}$, quantised; the charge and a column's coefficient as entries of the coupling matrix; the hand a pseudoscalar under the 48; a family's rows on the fan under the 48; a pair's arms in $\mathbb{Z}^2 \otimes \mathbb{Z}^2$ under the settings' integer matrices; and the six verbs are the only operations on them. The mass angle m of the frame is the rest pair's turn, $2\pi n/d$ per interval, a world's declaration and not the family's. Table S6 of the Supplementary Material lists every family of the register with its declared integers, its object, when it entered, the verbs that act on it and the click that reads it; a cell the tree does not name says so.

3 From click to click

Everything compared with nature is a count between clicks at a detector or a ratio of such counts; nothing inside the board is compared. This section is the chain from the ring's element read at a click to those counts and to the forms they take, the click frame under the hypotheses (A1) to (A3), each named where it enters; it uses the objects of Sections 4 and 5 by the names defined there. It opens with the six steps between two clicks, the two theorems of the Outside, and the one fact behind the uncertainty relation and Bell's excess; then the chain.

The steps between two clicks. Every comparison in this paper walks one chain of six steps, each with its kind (DETECTOR, GAMEBOARD, COMPUTATION, CONVERSION, HOST; Section 7).

- (1) The birth of a record's rows at a declared source: a declaration, the family's row of integers and the world file's source with its rate against its wall; the wheel's reading u is fixed here (Definition 2, B).
- (2) The flight: the six operations on $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ and the lattice, the phase turning per Link, the split and the rotation by the declared tables, the merge in the ring; the law (P3 the six operations, P4 locality, P5 one Link per interval, P9 the blind flight).
- (3) The arrival at a detector's Node and the click, the one measured event and an action of the law on the state, not a passive read (P6 one read-out, P10 the click axiom, P11 clicks only; the law's rule [20, section 10]): the record ends at the detector, its content moves into the detector's own record, and the click is stamped with the detector's own count; the flight of every other row is untouched, no back-action; the evaluation of the record at the roots of unity and the comparison (DETECTOR).
- (4) The detector's own count between two clicks, read under the key `clock_stamp` (DETECTOR; a key outside this paper's law identity, Section 5.1, named as such where it is read).
- (5) A ratio of two such counts, or a count itself (COMPUTATION on clicks; nothing of the board enters).
- (6) The identity the algebra states between the ratios, the form of nature compared with.

Step (6) is the only place where a formula meets a measurement: the click theorem gives the ratios' identity from the ring, Lorentz's factors under (A1) to (A3), Einstein's step and, in the limit under a shell average, Newton's form; a row of Table S1 compares the ratio of step (5) with that identity, and with nothing else. Anything read from the board itself, a tick, a presence mean, a body's own record, a shell mean, is a GameBoard diagnostic and enters no comparison; the failures of Table S1 are named by the step their earlier reading broke (Section 7).

Locality and discreteness Outside, two theorems beside P11. Two properties of the game above the board follow and are not assumed.

Theorem 2 (Locality Outside). *Under (A1), with the definition of Outside (P6 and P11: nothing leaves the board but clicks), every Outside event is a click and every click moves information one Node at most, so every Outside passage is a chain of clicks between neighbouring places, and the Outside inherits the Inside's locality through the conversion: no reading needs a jump, no velocity read exceeds one Node per interval of the tick, which in a detector's own count is $1/r_D$ Nodes per*

count, and no detector reads at a place no chain of clicks reaches. The one exception is the pair's click, one gather of one record from both settings, the law's one non-local operation.

Theorem 3 (Discreteness Outside). *By the click's definition, and under (A1) with the conversion, the counts of one detector form a free abelian monoid and their ratios lie in $\mathbb{Q}$: every Outside quantity is an integer count or a ratio of counts. The least separation of two places read is one Link; the least time a detector times by itself is a pulse to its neighbour and its return, two of its counts at $r_D = 1$; the least step of a moving record is one Node per k counts, so the Outside velocities are the ratios $1/k$, with the quantum $1/k - 1/(k+1) = 1/(k(k+1))$, finest near c and coarsest near rest; and c Outside is one Link per the least count, the bound at $k = 1$. Nothing continuous enters above the board.*

Both are the click frame's, its section 0 and its closing theorem [48], and Theorem 3 of Einstein Outside [24]; neither is an added assumption. What is more than (A1) is not a theorem: that the apparatus is itself inside the law, that emitters and detectors are records of the board and not a second kind of thing, is an assumption about the world or a programme for the law; this paper marks it so and does not use it.

The one fact and its two faces. One fact: a record is read once; the click reads one cell of one record and ends it (Definition 3; P6, P11). The uncertainty relation and Bell's excess are that fact seen from two sides; every claim here is the derivation's [22, section 22] or the click frame's [48, section 10], none new. The first face: a record is an element f of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its position the Node its rows end at, its momentum the turn per Link (de Broglie's $\lambda = h/p$ as the declared turn, Section 4.4); the click evaluates f at the N_ϕ -th roots of unity, the discrete Fourier transform on $\mathbb{Z}_{N_\phi}$, so position and phase are the two sides of one transform. On the identification of the circle as position and its transform as momentum, a hypothesis of the programme and not a derivation (Section 4.4), what holds exactly and with no limit is the support bound $|\text{supp } f| |\text{supp } \hat{f}| \geq N_\phi$ [71] (its equality cases at $N_\phi = 64$, by enumeration: one phase and all 64 roots, the antiphase pair and 32, the comb of every eighth phase and 8; COMPUTATION) and Weyl's relation $VU = \omega UV$ on $\mathbb{Z}_{N_\phi}$ [73, 74], from which $[x, p] = i\hbar$ follows as $N_\phi \rightarrow \infty$ with $\hbar = hN_\phi/2\pi$, Kennard's $\Delta x \Delta p \geq \hbar/2$ as that limit [72, 75], and at finite N_ϕ the entropic bound [76], the click's identity that the bits read and the bits not read of u sum to $\log_2 N_\phi$ per record. What makes it an uncertainty and not a spread is the one read-out: read at a Node the record gives its position and its phase vector is erased, read on the fan's angle it gives its momentum's direction and not the Node; the two are not jointly readable because there is one click per record, and the click acts (step (3) above); the reading is what is uncertain, not the board. The second face: a lamp releases one record with two arms, the joint labels 00 and 11 of weight 1, $\psi = (00) + (11)$ in $\mathbb{Z}^2 \otimes \mathbb{Z}^2$ tensored with its phase in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its coefficient matrix of rank 2, no $\mathbf{u} \otimes \mathbf{v}$ equal to it: entangled by construction; in flight each arm is local and nothing of one arm is written on the other; a setting is the integer rotation U_s with $U_s^\top U_s = n_s I$ (Definition 2, R); the pair's click is one gather of the one record from both settings at its completion, the joint pointer $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$, the cell's weight J^2 , the birth wheel selecting one of four cells through the ladder (Section 4.5). The rows are local and the read-out is not: the non-locality sits in one object, a record of rank 2 read by one gather at two clicks, with no signal between the arms; $E(a, b) = \cos(2\pi(a - b)/N_\phi)$ within the tables' rounding, the marginals exactly $1/2$ (Theorem 7; no-signalling an identity of the integers), $S = 2\sqrt{2}$ before the rounding and $176/64 = 2.75$ at $N_\phi = 64$ (Table S1, row 1a, against 2.42 ± 0.20 [90]), $181/64$ on the plateau (Theorem 8); Tsirelson's bound [86] a theorem of the operator algebra on the labels; not a hidden-variable value, because the pair's weight is one quadratic form with cross terms where a mixture is linear (Fine's theorem [85]). What links the two faces is the same one read-out per

record: it makes the support bound an uncertainty and denies Bell a hidden joint distribution. Not proven, and said so [22, section 22.3]: that Bell's excess is the same Fourier fact as the support bound. The standard names are the things compared with; the model's own words, the click, the record, the gather and the wheel, name the mechanism.

Inside and Outside, and how a click arrives at Lorentz. This paragraph and the two after it are the conditional extension of the law, the click frame under (A1) and (A2): where (A2) enters, the result is not a consequence of the law as built, which meets (A1) only, and each line names it; the r -free lines follow from (A1), (A3) and the conversion alone. A packet moved Inside, from place to place, and what it says Outside: Inside, the law propagates records as a beam by the algebraic formulas of Section 5.1, where no one measures; Outside, the game above the board, there are detectors and their clicks only, the beam passing to them with amplitudes at the click (P11). So a thing propagates Inside as a beam, rows on digital lines at one Link per interval, and shows Outside as a wave: what a detector reads is the click's weight, the quadratic form of the rows' phase counts (Definition 3's $X^2 + Y^2$, the same integer as the square of a phased sum), which adds and cancels as a phased sum at the roots of unity does, so Young's bands and the Mach-Zehnder's ports are the beam's rows read at the click and nothing else (Section 4.4); the wave is the read-out of the beam, not a second thing on the board. A detector's clock is its own record's count of what it emits and receives back, no Node holding a time and an open face without a body having none; a velocity is Nodes apart over counts apart between neighbouring detectors; the read-out is the click (Definition 3). The frame [48]: "To convert the lattice and the momentum inside it into motion Outside, the game above the board, one uses clicks that pass information, a click being the passage of information from Node to Node at most one Node per interval (A1) and its content an amplitude with a phase that splits each interval between staying and hopping, the mass the staying share (A2); Lorentz of the world is assumed, and what is shown is that this conversion brings it. The symmetry of the conversion, the transformations between click families that preserve (A1) and (A2), is the Lorentz group up to scale, acting on the amplitudes of the passing packet as the Dirac walk's covariance, so that the passage of clicks is Lorentz up to corrections of order m^2v^2 , exact in the continuum limit, a detector's own rate $\sqrt{1-v^2}(1-\kappa^2/6)$ with the mass angle m the spacing over the reduced Compton wavelength; in three dimensions the lattice's anisotropy begins at the fourth order." (A1) and the one line, the relativity of the two directions ($k_{AB} = k_{BA}$), are special relativity's two postulates in the click language, so what is arrived at is Einstein's road walked on clicks, and what the board adds is (A2), the amplitudes that carry $\sqrt{1-v^2}$ into a detector's own rate. The law as built meets (A1), one Node per interval with the flight blind to the phase (P9), and not (A2), its hop a whole-record schedule with no staying amplitude; so its clicks carry the group and its counter runs at the rest rate at every speed (rows 4a, 4b), the two one-way Doppler factors read through records apart by $1-v^2$, the measurable that decides. No wall of the law and no declared identity enter the reading; Eq. (6) is an Inside formula on the record, read by no detector, whose Outside formulas the conversion gives. What the system gives is a computation Inside whose passage Outside is the clicks, at the quantum level as well; how far Outside represents reality is what Tables S1 and S4 show, result by result.

The step beneath and the step above. The Inside step of this chain is not Eq. (1) but the click frame's walk, one amplitude per record that splits each interval between staying and hopping (A2), whose exact invariant is $\cos \omega = \cos m \cos \kappa$, ω the walk's frequency per interval and κ its momentum angle per Link; Eq. (1), which hops whole, carries W only as the declared identity of Eq. (6) (Table S2). The paper's centre is the step beneath the lattice, Eq. (6) in its frame,

the six operations on the record with W its invariant; the direction is one way: the step beneath is the axiom, the conversion of (A1) and (A2) is the map, and the step above is its image, a theorem named as what Einstein wrote. From W , by the conversion, the step above follows: Wc^4 is $E^2 = E_0^2 + p^2c^2$, the gate E_0/E' is $1/\gamma$, a detector's own rate, the drive's fraction is the click's velocity, and the two Doppler factors are the step above read at two detectors. The click frame's section 7 [48] shows the chain with no square declared: the walk's exact invariant $\cos \omega = \cos m \cos \kappa$ gives $\omega^2 = m^2 + \kappa^2 - m^2\kappa^2/3 + O(6)$; the law's own Planck map ($E = h_q n/d = h_A f$, $p = \hbar \kappa$ per Link, $E_0 = \hbar m$, the load identity $3\hbar n = Q S d$) gives $E^2 = E_0^2 + c^2 p^2$; the whole unit $E' = E/c^2$ gives $W = E_0'^2 + 3 \mathbf{p} \cdot \mathbf{p}$ with $3 = 1/c^2 = d$ exactly on the body diagonal (2.954, 2.971, 3.000 by direction, the root's rounding), the corrections of relative order $m^2\beta^2$, exact in the continuum limit. So W is shown to that order as the conversion's identity under (A1) and (A2), no square declared anywhere in the chain [26], and Einstein's relation follows from the Inside step under (A1) and (A2), to second order, an extension the law as built does not meet, its rows hopping whole without (A2); for the rows' dynamics, which hop whole and lack (A2), W remains the declared identity of [6, record 270], the reading's identity and not yet the rows'. In one chain, with nothing of Einstein's put in: the Inside step, the click frame's walk on the record with W its invariant (one amplitude per record that splits each interval between staying and hopping, (A2), whose exact invariant is $\cos \omega = \cos m \cos \kappa$; Eq. (1), which hops whole, carries W only as the declared identity of Eq. (6)); the conversion, an emitter and a click at one Node per interval (A1) and the amplitude's split (A2); the symmetry of that conversion, the Lorentz group up to scale (the click theorem); and its image, the Outside step, a detector's place apart and count apart, whose most general form above the board is Einstein's step as physics writes it, $d\mathbf{p}/dt = \mathbf{F}$ with $\mathbf{p} = \gamma m \mathbf{v}$, $E^2 = E_0^2 + c^2 p^2$, $d\tau/dt = \sqrt{1 - v^2}$, and the geodesic's first terms ([24], section 3). Lorentz of the world is assumed, through (A1) and the relativity of the two directions ($k_{AB} = k_{BA}$), and what the chain shows is that the conversion brings it: the Lorentz group up to scale is the symmetry of the click families that preserve (A1) and (A2), and Einstein's equations are the form of the Outside step, one step above the board; nothing here derives the Lorentz group from the six verbs, and "nothing of Einstein's put in" says that no factor of Einstein's enters the chain's steps, not that the symmetry is derived: the step's r -free lines (the round trip, the composition, the aberration) are Einstein's exactly from (A1), (A3) and the conversion alone, and its scaled lines are Einstein's if and only if $r^2 = 1 - v^2$, which (A2) gives to order $m^2 v^2$ and the law as built, at $r = 1$, does not. The same road serves Newton (below) and, in the same font, what it does not reach: length contraction, the perihelion's second order, the bending of light, waves, horizons and the field equations [24].

The smallest thing above. What the Outside step is made of follows from (A1) and the click's definition with no run: a click is a Node, a detector's own count and what arrived, and the counts of a family add, so their ratios lie in $\mathbb{Q}$ [48]. Hence the least distance a click can report is one Link, Nodes apart being an integer; the least time a detector times is one count, a pulse and its return two; the least step of a velocity at k counts is one Node per k counts; c is one Link per the least count; and the amount w of a row is the unit that arrives. Every reading Outside is a whole number of these, which is why the world above the board is quantized, a theorem of (A1), (A3) and the conversion (Theorem 3 of [24]; the velocity's quantum not read as such): physics above cannot take this from its own formulas, and the Inside gives it. The speed of light Outside is then the arrival count over the Euclidean distance, $c_D = N_l |D|/T_D$ by a row's age or by a pulse and its return, its anisotropy $1/c_D^2 = 2.954, 2.971, 3.000$ on the heading, the face diagonal and the body diagonal and $1/\sqrt{3}$ in the limit [23].

Newton from the small step. The same way, with five assumptions of the law as built and no (A2) (the crowd read as the age moment, the age wall at coefficient 1, the push, the flight blind, the reading a pulse and its return) and the spreading of K beams over a shell of $N(r)$ Nodes, which is arithmetic and not a law, the fall, the equivalence principle, Kepler's ratio and the clock in a crowd come out Outside. The inverse square comes at the shell mean's rung and the equivalence exactly, M_A cancelling record by record. The register's D3, T and X are their confirmation, and the inverse square's decisive reading is not made [48]. Every result Outside is thus the image of one Inside formula under the conversion, and Table S4 lists the map row by row with the reading that sits on the Outside side.

From an Inside formula to an Outside formula, family by family. The same passage, written three times in full with the attributes of the row tuple of Section 5.1 (Node, direction, age, phase, emitter, content per unit, record, label, amount, multiplicity) named at each step, so that a reader sees which attribute each formula reads and where the conversion takes it Outside. *Light through two slits* (the family **light**, $h = 1$, the circle turns): the lamp puts one record on the board with two rows, one per opening, each with amount 1, multiplicity 2, the lamp's phase p_0 at birth and its own direction on the fan; Inside, each row's phase advances by the declared turn per Link along its digital line, so at the pixel x the two arrive with the phases $p_1(x)$ and $p_2(x)$, the difference $\Delta(x)$ the path difference times the turn (the attributes read: direction, age, phase; the content plays no part, $h = 1$); the merge adds them, $f = x^{p_1} + x^{p_2}$ in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, and an equal pair at $\Delta = N_\phi/2$ cancels; the click evaluates f at the root and squares it, $R(x) = 1024(65536 + C[p_1]C[p_2] + S[p_1]S[p_2])$ (the formulas of Section 4.4) (the attributes read: amount, multiplicity, phase); the rung lays the pixels on the ladder and the birth wheel selects one, the record is deleted. Outside a detector holds only its counts per pixel: over N_ϕ births the count at x is $b_x - b_{x-1}$, proportional to $1 + \cos \Delta(x)$ within $1/N_\phi$, bright where the two ages differ by a whole turn, which is Young's law with the spacing set by the turn per Link; the reading is the bands' centres 23.5 pixels apart for the exact law's 23.3 (L2b, DETECTOR). *A massive body's clock and its fall* (the families **m** and **mass**, $h = 0$, no circle): a body carries its Node, its content M , its momentum $\mathbf{p}$, an integer 3-vector, and its counts table, each count an accumulator (s, r, d) ; Inside, its drive moves it at the pace $|p_a|/(N_l N_w M + |p_a|)$ per axis where the axes' fires do not coincide, one Link per interval in all (the attributes read: content, momentum), its clock counts at the rate $1/(1 + a_\tau n/d)$ with a_τ the age moment of the crowd's rows at its Node (the crowd's attributes read: age, amount, direction; the body's: its counts), and the push reads the label flow $\mathbf{a} = \sum \text{amount} \times \mathbf{u}_D$ of the arrivals into $\mathbf{p}$ (the attributes read: amount, direction, content). The conversion is a lamp on the body and a detector at rest: the lamp's births are the body's own counts, the detector counts their arrivals with its own count, and Outside there are only the two counts and the births' Nodes. So $1 + z$ is the ratio of the two counts. The ratio of two lamps' shifts at 3 and 6 Links is the clock's form at two distances, 1.907 for the pin 1.909 (series T, DETECTOR; read before the generic entry of 2026-09-22, not readable under the law as it stands, being re-read in the weak field). The fall is the birth's Node per count read at a line of detectors, the same Node on 138 of 139 births with the held mass four times (D3, DETECTOR), the equivalence read with no mass in the reading. *The pair* (the choosers **sa** and **sb**; the pair's record carries the labels $\{0, 1\}$ with integer weights on two arms): Inside, a setting a acts on an arm's label pair by the integer matrix U_a (the attributes read: label, amount, phase on each arm; the record's identity joins the arms), the joint weight is $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$ and the cell's offer $R = J^2$, Theorem 7; the conversion is the one gather of the record from both settings at the completion interval (P6), the rung over the four cells and the wheel's choice. Outside the two detectors hold four counts per setting pair: $E_{N_\phi}(a, b) = 4c_{++}/N_\phi - 1$ and $S(N_\phi)$ from them,

Theorem 8, with each party's marginal $N_\phi/2$ exactly; the reading is $S = 2.75$ at $N_\phi = 64$ and $32/64$ in every bin (series L, DETECTOR). In each of the three, Outside holds counts and Nodes and nothing of the attributes themselves: the phase, the amount and the multiplicity of one record never pass singly, the content passes as a wall's scale, and what passes is what the click compares. Table S4 of the Supplementary Material names, row by row, the family and the attributes each Inside formula reads.

What the Inside step gives Outside that the continuum cannot state. Formulas of the six verbs taken Outside by the conversion, each with the click that reads it and its kind; none of them is a formula of the continuum; the full lists, each candidate under the three tests with its reading, are [28] for quanta and bodies and [29] for light. (i) The quantum of the Outside step, stated above (one Link, two counts, $1/k$ with its quantum $1/(k(k+1))$; Theorem 3 of [24]), read as a least step in series S's face clicks 369 and 345 within two of the flight table's and D3's 138 of 139 births at the same Node (DETECTOR); the velocity's quantum and the pulse's two counts not read as such. (ii) The pair's sum at the CHSH labels is the exact rational the rung gives, $|S(N_\phi) - 2\sqrt{2}| \leq 8/N_\phi + 0.0444$, above the bound at $N_\phi = 16$ and 32 ($S = 3$), the plateau $181/64$ from 512 through 8192 below it by 3.02×10^{-4} and $5793/2048$ above at 16384 (Theorem 8; the frame's section 10); measured at seven N_ϕ (Figure 6, DETECTOR). (iii) The perihelion: the drive's pace departs from p/m_i at first order per axis, so a bound orbit's apsides move per revolution by an angle of order β with the lattice's symmetry, where Einstein has $6\pi\beta^2$ isotropic, larger than nature's by $1/(6\pi\beta)$, about 330 times at Mercury's pace if its coefficient is of order one, and of the wrong symmetry: FAIL on the law, pinned in order and symmetry, the coefficient NOT MADE; under the identity $\pi\beta^2$, one sixth; the deciding reading, the apsides of a lamp's orbit at two orientations to the axes, NOT MADE, its pin written before any run, an apsidal motion of order β that changes with the orientation and not with the mass held ([24], II.12). (iv) The anisotropy of c by direction, stated above (series Q, 290 of 290, DETECTOR; [23]). (v) The equivalence's constant: the clock in a crowd and the accelerated detector read one shift, $\delta k = (nS/d)(gY/c^2)$, Einstein's gY/c^2 exactly when $nS = d$ (times the grain's $\tau_{Lc} = 0.993$ on a heading, 1 in the limit of every direction), a condition on an Inside declaration and not a result; the register's clock and crowd worlds declare $nS/d = 16$ (GAMEBOARD, the world files), so the condition is not met by them, the form's readings are series T's 1.907 and series X's k (DETECTOR; T read before the generic entry of 2026-09-22, not readable under the law as it stands, being re-read in the weak field), and the constant's reading is NOT MADE ([24], II.10). (vi) Light's bending: on the law of this paper, the runtime before the generic entry, 0, the flight blind to the crowd (series K, 0.000 pixel, DETECTOR; FAIL against 1.75 arcseconds); under the key **optical**, the flight in the age wall's set with a declared c_f , the delay $c_f(nS/d)(GM/c^3)\ln(4r_1r_2/b^2)$ and the bending $\alpha = 2c_f(nS/d)GM/(c^2b)$, Einstein's at $nS = d$ and $c_f = 2$, NOT COMPARED, the route chosen ([24], II.11). The chain, algebra first and the run after (Section 4.3; Table S1, row 13): the step algebra's ring on the law as built, $C_{\text{ring}} = 5.12$ at $b = 6$, the M/b form recovered and 4 not reached (COMPUTATION [37]); under the one constant 3.65 to 3.92 against 4, $2c_f$ within two stated factors (COMPUTATION [38]); the run of 2026-09-22 under **flow_link**, the $\gamma = 0$ ring at $b = 6$ inside its four pins (DETECTOR) and, on the split ladder, every ring read inside its pins (the deciding pin 0.731 against 0.731 ± 0.025 , DETECTOR), the constant through the stated factors outside the one-grain pin by 2.40, 1.07, 1.40 grains at $\gamma = 1$: DOES NOT CLOSE within the grain [41]. The conjecture, labelled so: on the one constant the deflection coefficient is $2(1 + \gamma)$, γ the declared input, conjectured from the algebra and not read (Section 4.3). (vii) Dark energy's shape: the diagram of redshift against flight time read at a detector carries, beside the throw's Doppler $1 + z = 1/(1 - x)$, $x = H\tau$ (Milne's coasting

form, $q = 0$ exactly), one factor $g(\tau) = (1 + k_A(\tau))/(1 + k_B)$, the emitters' clock stretch by their crowd over the detector's own; with $g = 1 + g_1x + \dots$ the deceleration parameter a reader fits is $q_{\text{eff}} = -2g_1/(1 + g_1)$, an apparent acceleration for every $g_1 > 0$ with nothing accelerating on the board, a property of the conversion and not of any dynamics ([27]; rung 2, the throw's continuum). The register read that shape (series G, the accelerating form the nearest of three in all six pushing windows, DETECTOR) and the coasting throw's $q = -0.108$ in 0 ± 0.25 (G2, inside its pin; row 3, not predicted, the crowd's history missing). What the law does not give, stated in the same font: the size, $g_1 = 0.36$ for -0.53 , an input of the crowd (a declared pair, four orders above the known wells); the sign from its own accumulation of the crowd, $q_{\text{eff}} = 2k_0(3 - k_0)/(1 - k_0)^2 > 0$, the older light born in a thinner crowd; and the brightness of a standard lamp, which reads $q_{\text{eff}} = 2/(1 + g_1)$, a deceleration, one factor of $1 + z$ short of the expanding form because a click carries the content of its birth (row 11a, not predicted). The model has no cosmological constant and needs none as a rule; it states the supernova diagram as a failure, not a resolution. (viii) The atom. What the algebra gives first, on the law's integers (COMPUTATION [39]): the closing radius is an integer congruence of the action row, $\Delta A_a = 0 \bmod h$ on each axis with the three quotients summing to jN ; two closing radii stand as the square of two counts a detector reads, $a_j/a_i = (j/i)^2$ in the shell mean, checked by $T_j/T_i = (j/i)^3$, no root and no float; and a theorem of the click says that from a closed circle one click reaches only the loops with $2i^2 \geq j^2$, while a release of content unbinds, so the transition Bohr reads as one line is not an operation of the law: NEEDS A NEW RULE, named under its own identity (atom-level-v1, the release at a closure of the difference of two closures, the virial form, [40]). Stated as a conjecture, labelled so and not read [39, 40]: every closed loop with j circles per return has, in the shell mean, $a_j = j^2 h^2 / (4\pi^2 \kappa m)$, so $a_j/a_i = (j/i)^2$ and $T_j/T_i = (j/i)^3$; the loop's constant $-\kappa/(2a_j)$ on the virial form, $E_j = -\Delta A/(2NT_j) = -jh/(2T_j)$, is proportional to $1/j^2$ in the limit, so two lines would stand in the ratios $(1/i^2 - 1/j^2)$, 27/20 for Balmer's β over α , free of every declared constant; the conjecture is that the law's closed loops form Bohr's ladder in the shell mean's limit, conjectured from the algebra, not recovered. What is read: the $r = 12$ loop's staying under the centred step and its tightening (DETECTOR); no level and no line. The atom enters the paper by the algebraic formula alone, and the paper says plainly why: its levels cannot be read on any lattice that can be run. In nature the electron's orbit radius over the nucleus's radius is about sixty thousand (the Bohr radius 5.3×10^{-11} m over the proton's 0.84×10^{-15} m), while the lattice worlds put the electron at $r = 3$ to 12 Nodes from the proton, where the fan's grain (about 17 degrees per shell at $r = 3$, the flux 17 percent below the inverse square) is the named limit; a lattice of order fifty thousand on a side would be needed and is not run ([6, record 997, pending merge]). So the ladder is the algebra's convergence to Balmer's form in the limit: computed from the algebra, not read on the board; a lattice able to hold the scale is not run; not shown in nature. No run of the levels is cited as a reading, and row 6 stays NOT YET on the line. The run after it: the hydrogen loop at $r = 12$ (series H's world, the electron's held content on the proton's push) escapes under the drive as built at 3407 intervals (DETECTOR), the law's own FAIL, beside which the key's PASS below stands. The cause, read from the baseline's records by algebra [33]: the count primitive fires a body's step when its accumulated motion reaches a whole wall, so the body's Node lags its accumulated motion by half a Link in the mean on each axis, toward the motion's sign, and the push read at the lagging Node has a part along the motion, $(S_1(\mathbf{v})/|\mathbf{v}|)/(2r)$, about 5 percent of the radial at $r = 12$, which pumps the loop's angular momentum and energy (up 30 and 45 percent over the first three quarters of a turn) until the loop opens. The centred step (`centred_step`, a declared start of the drive under its key, off by default, every registered world byte identical without it: the count of a body's step the nearest whole number of its accumulator in units of the wall, the lag zero in the mean) is a hypothesis under its own name until the author makes it the law; under it, in the one run of 2026-09-22 [32], the loop stays: no escape face click in 7500 intervals,

23 quarter crossings (C1 PASS, DETECTOR), tightening from 12 to 7 Links at two crossings over five turns, the period falling from 1431 to 1040 (C2, C3 FAIL as declared, DETECTOR), no pin moved after the run; no line of Bohr's is read, and none is computable by a run of the law as built, no rule of the six quantising a body's loop (Table S1, row 6). The families of the register, each with its declared integers, its algebraic object and the click that reads it, are Table S6.

The limits, and what does not return. The content and $\sum w^2/\mathfrak{m}$ are conserved exactly, the offered norm only within the tables' rounding. The covariant readings [6, record 270] read one Inside formula Outside on one axis, series S against its pins: a body's record carries the exact square of its energy,

$$\boxed{W = E_0'^2 + 3 \mathbf{p} \cdot \mathbf{p}, \quad E_0' = E_0/c^2 = N_t N_w M, \quad c^2 = 1/3} \quad (6)$$

compared and never rooted, with E' the largest integer whose square is at most W and every count of the body gated by E_0/E' ; multiplied by c^4 it is the energy-momentum relation of special relativity (series S, Table S4, inside their pins). Eq. (6) is a declared identity of the record's representation in the law as built, not derived from the rows' dynamics; under (A2), an extension the law does not meet, W is shown to second order (the step above, this section). The hypotheses beside the law are on the tree [14].

4 From the algebra, the physics

What follows from the algebra is stated first in one paragraph, each item with its word (exact; recovered in a limit; reached under named hypotheses; conjectured), and then subsection by subsection with the proofs, the runs that read each form after a detector, and the words kept apart.

What is proved. Derived Inside, exact on the GameBoard, no experiment needed: the group of the six Ports and, within it, the group of order 24; the books' conservation and the lattice form of the continuity equation; Gauss's law of a free family's flux under its world condition; the split's isometry and the injectivity of the interval on the amplitudes for a fixed event history; the pace of every direction, isotropic within $1/T_D$; the click's weight a positive quadratic form of power 2; the exact marginals of a pair; the CHSH sum as an exact function of the grain and the tables; Planck's and de Broglie's relations as identities under the declared dictionary. Recovered in the limit of the grain, with a rate: Born's rule to $1/N_\phi$ per cell and the cumulative rungs to $1/(2N_\phi)$ with the exact tables (to the tables' rounding with the declared ones, Section 4.4), Tsirelson's bound to $8/N_\phi$ plus the tables' 0.0444 ($2\sqrt{2}$ in the joint limit of N_ϕ and N_t), Young's spacing, the attained value of c , the continuity equation's differential form. Recovered under an average that covers the shell, with the condition in the claim: Newton's and Coulomb's inverse square, the retarded potential and Poisson's equation, the clock's $1/r$ form. Each names the known form the algebra recovers (Gauss's, Newton's, Coulomb's, Born's, Tsirelson's, Planck's, Einstein's); the paper's own are the integer road, the finite- N_ϕ values (Born's bound $1/N_\phi$, the plateau 181/64, the fixed-table limit), the quadratic read-out characterization on $\mathbb{Z}_{N_\phi}$ (Gleason-like), the measurable $1 - v^2$ and the entropy identity of the click (an identity of the wheel); the value $1/\sqrt{3}$, Tsirelson's limit and the dictionary's identities claim no novelty. Conjectured from the algebra and not read, labelled so wherever they appear (Sections 4.3 and 7.2, Table S5): the bending's coefficient $2(1 + \gamma)$ on the one constant, and the atom's ladder $a_j/a_i = (j/i)^2$, $E_j \propto 1/j^2$, with Balmer's 27/20 in the shell mean's limit; the ring's run reads its rule inside its pins and its constant outside the one-grain pin,

and the ladder is computed from the algebra, not read on the board. After a detector every reading sits inside an expectation written before the run; against nature, of the ten observables the law as built predicts (eleven once row 14 is read), four agree within tolerance (1a, 2b, 9; 5a as a bound met), six disagree (1c, 6, 7b, 8a, 8b, 13), and eight are not predicted by construction (3, 4a, 4b, 5b, 7a, 8c, 11a, 11c), each with its missing piece named in its cell and its kind, the instrument missing in the model, nature's number a bound or a fitted parameter, no click definition yet, or the model's own number a declared input; 2a NOT COMPARED until its source is verified; six rows keep their words (12 history, 2c not compared, 2a as said, 10 not yet, 11b a pin, 14 not yet read against nature), and 1b is a control, not counted; the hypotheses beside rows 6, 13 and 14 change no verdict of the law (Table S1). What no rule of the law forces, with the check that decides each. Newton's inverse square and Poisson's law closed after a detector (G not read; D3): decided by a closed orbit's period at two radii under the directional drive (form B, built off by default, series Y), the escape's count at a face detector and the clock's form at more distances. The Lorentz factor, the law meeting (A1) and not (A2): decided by the ratio of the two one-way Doppler factors through records, $1 - v^2$ under the law and 1 under Lorentz, and by the split of a row between staying and hopping (A2), a hypothesis not opened. The bending of light, the flight blind to the crowd: decided by a rule under which a row's wall reads the crowd's age moment (a hypothesis, its coefficient an input) against nature's 1.75 arcseconds. The masses' values, every rule linear in the content and every equal split a fixed point: decided by a nonlinear closure on the amounts under its own identity, the family table an input until then. The click's harmonic constants and the circle as position (a hypothesis, Section 4.4): decided by the least-rank click, an axiom of the apparatus pinned by the two-slit period and Malus at 22.5 degrees, and by the single opening against its pin (row 10). And a CHSH measurement at 10^{-4} , deciding 181/64 against $2\sqrt{2}$.

4.1 Geometry and symmetries: the directions and $c = 1/\sqrt{3}$

The six Ports and their group are Theorem 1 of Section 2; the symmetries of the law and the pace follow from them.

The symmetries. The law is invariant under the translations of the box and of the interval, and under the 48 signed permutations of the axes, the group of the octahedron of the six Ports and of the cube ($3! \times 2^3$; the 24 of determinant +1 its rotations, the 24 of -1 its reflections, told apart by the hand, a pseudoscalar of the group), up to the law's two declared ties, the digital line's axis order and the collision's Port order ([2], note 39 and section 4); under the circle $\mathbb{Z}_{N_\phi}$ exactly on the turns between events and at the click only up to the tables' rounding; and under the permutations of the families, each with its columns of the family table carried along [9]. A boost, which mixes a step in space with a step in time, is not among the 48: Lorentz's symmetry is not a symmetry of the GameBoard but of its clicks Outside, under the amplitude split (A2) that the law as built lacks, to second order in the velocity, the lattice's corrections from the fourth (Section 3). The 48 are the GameBoard's stand-in for the rotation group; angular momentum is conserved only up to the GameBoard's grain, and no Noether theorem for the map is claimed [6]. The name of the program counts them.

The pace of the rows, and the octahedron. Two rules are the whole kinematics of a row: at most one Link per interval (the locality of the GameBoard) and the digital line of its direction, so that after one period of $S_1 = |D_x| + |D_y| + |D_z|$ Links its displacement is D itself (straightness). The octahedron $|x| + |y| + |z| \leq 1$ of the six neighbours (Figure 3), scaled by the age, is the causal front, inside which every row stays; a row's mean Euclidean pace lies within $1/T_D$ of its inscribed sphere

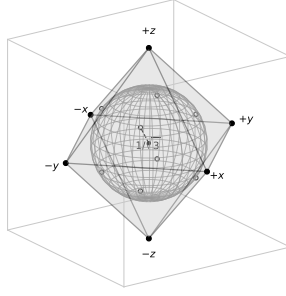


Figure 3: The Nodes one interval from a Node: the six neighbours are the vertices of the octahedron $|x| + |y| + |z| \leq 1$, the causal front of one interval; its inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the cube diagonals and is the rows' pace c (Proposition 1); the faint cube shares the 48 signed permutations of the axes. Drawn by `octahedron.py` from the definitions; no run.

of radius $1/\sqrt{3}$, which touches the faces on the cube diagonals (Proposition 1), so the vertex's pace, one Link per interval along a heading, is reached by no row's mean pace: the front is a bound and not a signal.

Proposition 1 (The pace). *For every primitive direction D , $S_1 N_l \leq T_D$: the Manhattan pace $S_1 N_l / T_D$ is at most one Link per interval on every digital line, and $S_1 = \sqrt{3} |D|$ exactly when $|D_x| = |D_y| = |D_z|$. A pace v that is the same in every direction and at most one Link per interval on every line satisfies $v \leq |D| / S_1$ for every D , hence $v \leq 1/\sqrt{3}$, the radius of the octahedron's inscribed sphere; and the table's Euclidean pace obeys $1/\sqrt{3} \leq N_l |D| / T_D < (1/\sqrt{3})(1 + 1/T_D)$ for every D .*

Proof. $S_1^2 = (|D_x| + |D_y| + |D_z|)^2 \leq 3(D_x^2 + D_y^2 + D_z^2)$ is the Cauchy–Schwarz inequality against $(1, 1, 1)$, an equality exactly when the three sizes are equal; so $S_1 N_l \leq \sqrt{3|D|^2 N_l^2}$, and since $S_1 N_l$ is an integer, $S_1 N_l \leq \lfloor \sqrt{3|D|^2 N_l^2} \rfloor = T_D$. A row at the Euclidean pace v on the line of D makes S_1 Links per length $|D|$, so $v S_1 / |D|$ Links per interval on average, and at most one Link per interval bounds this by 1: $v \leq |D| / S_1$. The least of $|D| / S_1$ over directions is $1/\sqrt{3}$, on the cube diagonals ($1/\sqrt{n}$ in n dimensions). $T_D \leq \sqrt{3} N_l |D| < T_D + 1$ gives the pace's interval. $\square$

So $c = 1/\sqrt{3}$ Links per interval is the largest isotropic pace the two rules allow, and the table sits at it; that it sits at the supremum and not below is the flight's declared wall $T_D = \lfloor \sqrt{3|D|^2 N_l^2} \rfloor$, the design's third statement, a rule chosen among few (P9; the derivation's 24.1, rows 7 and 9), not a consequence of the two; given the wall the value is the flight operator's norm, so the value rests on locality, straightness and the declared wall. Stated exactly: the two rules give $c \leq 1/\sqrt{3}$ (one Link per interval allows the pace 1 on a heading and only $1/\sqrt{3}$ on a cube diagonal, three Links for the displacement $\sqrt{3}$), and what is derived is the largest uniform pace the GameBoard allows under them; why the rows realise it, and what a detector in motion reads of it, are not derived here (Section 3). The anisotropy is the table's rounding, below $1/T_D$ in every direction and so of order $1/(\sqrt{3} N_l)$ (Table S1, row 5a, the bound on N_l ; the bound assumes the rounding's anisotropy is of order $1/N_l$ at every N_l , as computed at 64; an upper bound alone gives a sufficient N_l , not a necessary one); nothing measured enters the value. The number is the bound of the Courant condition for the wave equation, $c \Delta t / \Delta x \leq 1/\sqrt{n}$ [55], by this same argument, and the lattice

Boltzmann sound speed $c_s^2 = 1/3$ [56]; the paper adds the exact integer table at the bound and its rounding's anisotropy as a bound on the grain, and claims no novelty for the number. Stated with its ground: the pace of a direction, $N_l|D|/T_D$, isotropic within $1/T_D$, and the Manhattan bound $S_1N_l \leq T_D$ are exact in integers; the isotropic bound $1/\sqrt{3}$ is a theorem about a pace the same in every direction, attained on the eight diagonals, above on every other registered direction, and a limit of the grain N_l .

What a detector measures. Two registered runs read the pace after a detector (the Supplementary Material, S3): series Q, a lamp releasing one row on each of the 290 primitive directions with $|a| + |b| + |c| \leq 6$ toward detectors on the six faces, every click at the derived interval, Node and face, the pace 0.5718 to 0.5893 Links per interval at finite ages against the table's asymptotic 0.5774 to 0.5818; and series L7, the cone, a row of 17 Links on an axis and a row of 24 on the plane diagonal, the same Euclidean distance to one percent, at their counters at the same age.

4.2 Conservation, the flux and the forces

The books. Every message released is on exactly one line of the books at every interval, released equals in transit plus absorbed plus escaped plus cancelled, per family, in amount and in content, exact at every interval and at every operation (P3, P5): the carry is a bijection $s \mapsto (e, s - ed)$ that loses nothing, and the split and the rotation are integer matrices whose conjugate transposes invert them (Theorem 4). From the books the continuity equation of the content follows exactly on the GameBoard (the derivation's 25.4 [22]); it is the first row of the ledger. A paid message carries its label out of its emitter at birth and into its reader at its end, so for paid messages momentum is conserved and the third law holds message by message; a free message, a field's row, takes no recoil at birth and pushes its reader by its label times the reader's content. For a free family the third law is a symmetry of two readers, body A pushed by B's field and B by A's, the two pushes $-M_A\mathbf{a}_B$ and $-M_B\mathbf{a}_A$ (M the content, $\mathbf{a}$ the acceleration the other's field gives) equal and opposite, and not a balance per message; the momentum lines of the free books are a report of the GameBoard, and the momentum of the bodies and the field together is not booked as conserved (the derivation's 9.1 and 12.4 [22]).

The operations. Six operations, of three of Section 5.1's kinds (the translation, the bilinear form, the Euclidean division), make the whole of gravitation and electricity in the law. The free release: a body of content M_B holding a free family releases at each self-creation the carry of the count of rate $M_B\eta$ against the wall 1, η the release rate per unit of content per direction, on every declared direction, so a fan of K directions releases $q = KM_B\eta$ per interval. The walk: each unit walks its digital line at the pace of Proposition 1. The reading: the label flow $\mathbf{a} = \sum \text{amount} \times \mathbf{u}_D$ of the rows arriving at the reader's Node, $\mathbf{u}_D$ the integer unit vector at the scale N_l ($|\mathbf{u}_D| = N_l$ within 1.35 percent on every direction). The coupling, a bilinear form with a declared integer matrix: per axis $\text{push}_A = M_A(\rho_A\rho_B - 1)\mathbf{a}$, the gravity column $-M_A\mathbf{a}$ and the charge column $+\rho_A\rho_B M_A\mathbf{a}$, ρ the declared charge per unit of content. The momentum: $\mathbf{p}_A \leftarrow \mathbf{p}_A + \text{push}_A$. The step, the Euclidean division: one Link per $d_p/|p_a|$ self-creations on the axis a , the wall $d_p = N_l N_w M_A + |p_a|$, N_w the world's width, so that for a body whose clock owes nothing the speed on the axis is $|p_a|/(N_l N_w M_A + |p_a|)$ per interval.

Gauss's law, exact. The walk is a translation: a free unit released inside a closed surface that no periodic axis crosses, with no absorbing reader inside it and within the unit's lifetime, crosses the surface outward once more than inward (a non-convex surface it may leave, re-enter and leave

again, the signed crossings summing to one, the crossing counted with the sign of the Port's outward normal; no splitter, absorber or cancelling pair inside the surface; no collision acts on a fan or on lone beams), so once the front has passed, the net amount crossing the surface per interval equals the release inside it for a constant release, exactly, and for a varying release the crossings at the interval t equal the releases at the retarded intervals, direction by direction. The crossings of an interior surface are the host's count (GAMEBOARD, a diagnostic); a face's border clicks are a detector's (DETECTOR). Proved from the definitions under those conditions; no detector reading of it is registered.

The inverse square as a shell mean. On a beam the flow per Node is the same at every Node of the digital line: along a beam the field falls as $1/r^0$, and off every beam it is 0, by the walk. The inverse square is the density of beams over a shell. A shell at distance r of unit radial width holds $\mathcal{N}(r)$ Nodes, and a beam of direction $\mathbf{D}$ crosses it at $S_1/|\mathbf{D}|$ of them (one Link per Node, $|\mathbf{D}|/S_1$ of Euclidean advance per Link), each reading the beam's arrivals, so the mean over the shell of what a body reads is

$$\langle a \rangle(r) = \frac{q N_l F}{\mathcal{N}(r)} \rightarrow \frac{q N_l F}{2\pi r} \text{ on the plane,} \quad \frac{q N_l F}{4\pi r^2} \text{ in space,} \quad F = \langle S_1/|\mathbf{D}| \rangle_{\text{fan}}, \quad (7)$$

as $\mathcal{N}(r) \rightarrow 2\pi r$ and $4\pi r^2$, with F the fan's Manhattan factor, 1 on a heading, $\sqrt{3}$ on a body diagonal, 1.287 in the mean over the registered fan [36], $4/\pi$ for a fan uniform in angle on the plane and $3/2$ for one uniform in solid angle in space, a factor that vanishes neither with r nor with the fan's grain (the presence, Eq. (10), carries the dwell $T_D/(S_1 N_l)$ per Link that cancels it into τ_L ; flow-link-v1, rows 13 and 14 of Table S1, divides every push by the same mean; the derivation's check of the shell mean on the six headings, series C, has the factor 1 on every heading and does not test it, and no non-heading fan check of the coefficient is registered); the departure at finite r is the GameBoard's count of Nodes on a shell, Gauss's circle problem, $\mathcal{N}(r) = 2\pi r + O(r^\theta)$ with the exponent $\theta \leq 131/208$ [69], so the shell mean's relative ripple vanishes as $r^{\theta-1}$. No detector reading of the shell mean is registered; $\mathcal{N}(r)$ is the GameBoard's count of Nodes on a ring, a number of the derivation and not a reading. Derived from the update rules implemented in the simulator, in the shell mean, with the ripple named; on one Node the inverse square exists only in the limit of every direction, and a fan declared as a cube of directions has in that limit a cubic anisotropy of the declaration, $3^{3/2}$ between a corner and a face centre, absent for a fan declared within a ball.

Newton's law and the place of G . A body A of content M_A at the mean flow $\langle a \rangle$ takes per interval the push $-M_A \langle a \rangle$, and its speed on an axis is $|p_a|/(N_l N_w M_A + |p_a|)$; for $|p_a| \ll N_l N_w M_A$ the acceleration is push/ $(N_l N_w M_A)$ Links per interval squared, so

$$a = -\frac{GM_B}{r^2}, \quad G = \frac{K \eta F}{4\pi N_w} \text{ in space;} \quad a = -\frac{G' M_B}{r}, \quad G' = \frac{K \eta F}{2\pi N_w} \text{ on the plane,} \quad (8)$$

in Links cubed (squared) per unit of content per interval squared: G is the source's release rate per unit of content per direction, η , times the number of directions K , over 4π and the world's width N_w . No detector reading of either form is registered; the decisive reading, a closed orbit's period at two radii, is NOT MADE. Nothing of A enters: the equivalence principle is exact (the step rule divides by M_A). The third law at rest, the two bodies' pushes equal and opposite, holds to the grain of the apportioning: in series C, item 2, the two sources' momenta agree to six parts in 10^6 , a GameBoard reading, a diagnostic (the derivation's 9.1 [22]); it is a symmetry of the two readers and not a balance per message, the free release taking no recoil. Two departures from Newton's

law as written, both stated: the field is retarded at c (the front at r arrives at the interval 1 plus the flight table's first age at r Links on the heading), so the limit is the retarded inverse square and not action at a distance; and the speed the push builds, per axis where the axes' fires do not coincide and one Link per interval in all, saturates at one Link per interval, above the rows' pace, so a body can outrun its own field's rows on the drive as built. Derived from the update rules in the shell mean, G named, the width N_w an input, a choice of units (Table S5).

Coulomb's law and the one constant. The charge column adds $+\rho_A\rho_B M_A \mathbf{a}$ per axis; with $q_A = \rho_A M_A$, $q_B = \rho_B M_B$ and $\langle a \rangle$ proportional to M_B , the electric push in the shell mean is

$$\text{push}_e = q_A q_B \frac{K \eta N_l F}{4\pi r^2}, \quad a_e = \frac{q_A q_B}{M_A} \frac{G}{r^2}, \quad (9)$$

Coulomb's inverse square with the same constant as gravity, $k_C = G$ in the law's units, repulsive for like signs; the ratio of the two forces on one body is $-\rho_A\rho_B$ at every r . Derived under the same average as Newton's; no detector reading of it is registered. Nature's ratio for two protons, 1.24×10^{36} , is ρ_p^2 here, $\rho_p = 1.1 \times 10^{18}$ units of charge per unit of content: the hierarchy is the declared ρ , an input, not a formula of the law.

The ground of these derivations. Gauss's law of the flux and the equivalence principle are exact on the GameBoard; the third law at rest is the two readers' symmetry, to the apportioning's grain (a GameBoard reading, six parts in 10^6). The inverse square, Newton's and Coulomb's, is a limit under an average that covers the shell, in one of three forms: the shell mean (the ripple $O(r^{\theta-1})$, $\theta \leq 131/208$), the fan dense at one Node for a fan declared in a ball, or the mean over one closed turn of an orbit, the only one readable after a detector, which equals the ring mean at one radius of the loop, $\langle r^2 \rangle_\theta / \langle r \rangle_\theta$ on the plane and $\sqrt{\langle r^2 \rangle_\theta}$ in space ($\langle \cdot \rangle_\theta$ uniform in angle), exactly in the continuum for every closed turn by the area rule of a radial push ([47], Theorem B; Kepler's second law the comparison, not an input); against the ring mean at the angle-mean radius $\bar{r}$ the orbit's mean is smaller by $\bar{r}^2 / \langle r^2 \rangle_\theta$, between $1 - e^2$ and 1, equal only on the circle; on the lattice it holds up to four grain terms, the ring's count against $2\pi r$ (0.90, 0.95 at $r = 12, 24$), its incidence against the fan's L1 factor (1.07, 0.97), the drive's anisotropy (0.069) and one Node's dwell per crossing of a line (1.55 at the registered pace; its mean over the fan's crossings small, assumed), GAMEBOARD by formula; on one Node beside a source the field is a comb of beams, the average part of the claim (the ledger's caption). After a detector the equivalence principle is measured (series D3) and the $1/r$ force's scale symmetry reads consistent, 1.997 for 2.00 ± 0.18 , on loops that are not similar figures, resting on the $1/r$ push's scale symmetry and not on the equality above; the turns' $e = (r_{\max} - r_{\min}) / (r_{\max} + r_{\min})$ is not on record, the run's extremes bound it by 0.96 and 0.70 at $r = 12, 24$ (floors 0.07, 0.51 on the factor, GAMEBOARD by formula), which decides nothing ([47], section 5); the inverse square's decisive reading is NOT MADE and G 's value is not read; Coulomb's run is not made; Poisson's interior is read after a detector (series X, Section 4.3).

4.3 The delay field: the retarded potential, Poisson's equation and the clock

The rows of Table S5 on the delay field are written out here from the derivation's section 5 [22]; the register's series E reads probes on the GameBoard, diagnostics this paper does not carry, and the field's one measurement after a detector is the clock's form at two distances, series T (row 12 of Table S1; read before the generic entry of 2026-09-22 [35], not readable under the law as it stands, being re-read in the weak field).

What is delayed. Nothing in flight is delayed under the law of this paper, the runtime before the generic entry (P9): a row moves by the flight table at one pace, blind to the crowd (series K, Table S1). What a crowd delays is a clock. A body reads at its Node the presence of every row of another number (the engine's default) or, on an entry that reads the age, their age moment (the clock's word, P9) and owes after each self-creation the whole part of $a_r n_c/d_c$ intervals before the next, a_r what it read and $[n_c, d_c]$ the clock's pair of Section 5.1 ($[1, 1]$ in the scalar world, $[1, 2]$ in the age world), so that its rate is $1/(1 + a_r n_c/d_c)$; its push reads the flow of the same rows.

The two fields of one stream. Let a source release $q(t)$ rows per interval on a fan of every direction (the limit of Section 4.2's shell mean), each row moving at c and carrying its age. The presence Π at the distance $r = |\mathbf{x}|$ from the source at the time t is the rows dwelling there, and the age moment $\mathcal{A}$ is the presence times the age they carry, r/c :

$$\Pi(\mathbf{x}, t) = \frac{q(t - r/c) \tau_L}{4\pi r^2}, \quad \mathcal{A}(\mathbf{x}, t) = \frac{q(t - r/c) \tau_L}{4\pi c r}, \quad (10)$$

τ_L the radial residence factor, the intervals a row spends per unit of Euclidean distance, $T_D/(N_l|D|) = 1/v_D$ (1.72 on a heading, $\sqrt{3}$ in the limit of every direction); the intervals per lattice Link along a row's staircase are $T_D/(S_1 N_l)$, a different factor (1 on a body diagonal), and a finite shell's coefficient depends on its digital geometry, so the coefficient of (10) is the heading's on a beam and the limit's on the sphere. The presence is the retarded flux of a conserved stream; the age moment is the retarded potential of the source, since $f(t - r/c)/(4\pi r)$ is the retarded Green's function of the wave operator, so $\mathcal{A}$ obeys the inhomogeneous scalar wave equation with the release as its source,

$$\frac{1}{c^2} \frac{\partial^2 \mathcal{A}}{\partial t^2} - \nabla^2 \mathcal{A} = \frac{\tau_L}{c} q(t) \delta^3(\mathbf{x}), \quad (11)$$

δ^3 the three-dimensional Dirac delta, and in the static limit Poisson's equation $\nabla^2 \mathcal{A} = -(\tau_L/c) q \delta^3(\mathbf{x})$, solved by $\mathcal{A} = (\tau_L q / 4\pi c) / r$. The moments are sums over rows, so the field of many sources is the sum of their fields: the theory is linear. The flow the push reads is the stream's flux vector, $\mathbf{a} = (q N_l / 4\pi r^2) \hat{\mathbf{r}}$, $\hat{\mathbf{r}}$ the unit vector from the source and N_l the label each unit carries, which is $-(N_l c / \tau_L) \nabla \mathcal{A}$ in the static case; the gravity column reads $-M_A \mathbf{a}$ (Section 4.2), so a body's acceleration, the push over $N_l N_w M_A$ (the drive's conversion of a push to a pace, as in Eq. (9)'s a_e), is $+(c/(N_w \tau_L)) \nabla \mathcal{A}$, up the age moment's field, toward the source. Derived from the update rules, in the limit of every direction: the age moment is the retarded potential of the release and obeys the wave equation there. After a detector the interior of a shell reads flat within the tolerance, $k(2)/k(4) = 1.0029$ for the map's pin 1.0039 (series X, DETECTOR), the shell theorem's interior; the finite- r ripple is MEASURED ONLY.

The clock's redshift. A clock at the age moment $a_r = C_a M / r$, C_a the constant of Eq. (10) times the clock's pair, runs at $1/(1 + a_r)$; general relativity's clock at the potential runs at $\sqrt{1 - 2GM/rc^2} = 1 - GM/rc^2 - \frac{1}{2}(GM/rc^2)^2 - \dots$. Under the calibration $a_r = GM/rc^2$, an input of the dictionary (by the derivation's 5.1 and 3.3 the clock's constant and Newton's G have the exact ratio of the suspension pair times the width, so the calibration is a constraint on the inputs), the two agree at first order: the redshift between two clocks, $1 - a_r(r_1) + a_r(r_2)$, is the $1/r$ law. Derived at first order, the calibration an input. At second order the GameBoard reads $1 - a_r + a_r^2$ against $1 - a_r - a_r^2/2$, and its rate never reaches zero at a finite a_r : a different law in the strong field, no horizon (the crowd only slows; nothing stops a clock). No detector reading of the strong-field form is registered; the weak-field confrontation ($a_r \ll 1$) is a run the register

names as the follow-up, neither pinned nor made. Which word the clock counts, the presence or the age moment, was put to the GameBoard as series T, the pins written before the run [4, 16]: a lamp at rest, two sources at 3 and at 6 Links, the detector at $x = 110$; under the presence word the detector reads $1 + z = 1.3000$ at both distances; under the age word 2.6517 and 4.1500, the ratio of the two a_τ being 1.907 for the lines' pin 1.909, nature's 2.00 being 4.6 percent away; every pin met, measured and replicated [8]; read under the law before the generic entry of 2026-09-22; not readable under the law as it stands; the four worlds are being redeclared in the weak field (the pair [1, 16384], the crowd's age moment at most 1024 per Node on the light's path) and re-read, the new numbers to replace these; the gravitational redshift's ratio is a reading of the law before the entry, not of the law as it stands. What the detector establishes is the form at two Nodes, a lattice-exact reading; the continuum's $1/r$ is its limit under the average that covers the shell, and the reading is evidence for it, not its proof. The age moment is the clock's word, an assumption of the law (P9; [6, record 394]), not a derivation; the law as built counts the presence by default and the age moment on an entry that reads the age; the presence word is the alternative refuted on the form.

Light: no optical metric. A row reads nothing of the crowd: beside a mass it is neither bent nor delayed (series K, three worlds, the deflection 0.000 pixel and the delay 0.00 interval, DETECTOR; FAIL against nature's 1.75 arcseconds, Table S1); the law as built here is the law before the generic entry of 2026-09-22 (on main at 5b855791 since that day [35]), and the paper's readings under the entry are the ring's alone (row 13); the delay field is a metric for clocks and for the bodies' pushes, not for the flight (Section 3, the bending under the key). What the algebra gives, on the law's integers and before any run (COMPUTATION): under the key `optical` the walk of a light row past a held mass by the law's steps, read at a ring of Nodes at the impact distance b , recovers the M/b form, $C_{\text{ring}} = abc^2/(GM) = 5.12$ at $b = 6$ (5.44, 5.86 at $b = 3, 8$), flat in b within 15 percent and not 4 within the ring's grain, the factor left the fan's $F_{L1} = 1.421$ of the flow (3/2 in the isotropic limit) [37]; with the flow counted per Euclidean Link of its line (flow-link-v1, the one constant: the push's constant equal to the clock's, 0.990 against 0.993) the ring reads $C_{\text{ring}} = 3.65, 3.82, 3.92$ at $b = 6, 3, 8$ against 4, that is $2c_f$ within the finite path's $L/\sqrt{L^2 + b^2}$ (0.974 at $b = 6$) and the shells' 0.990, one to three grains; $4 = 2(1 + \gamma)$ with γ an input read back, not derived [38]. Stated as a conjecture, labelled so and not read: on the one constant the law's deflection coefficient is $2(1 + \gamma)$ with γ the declared input, the M/b form with Einstein's factor 2 at $\gamma = 1$, conjectured from the algebra (the step algebra's simulation of the ring, 3.65 to 3.92 against 4, closing to $2c_f$ within the two stated factors to one to three grains, COMPUTATION), not recovered and not read: what is read is series K's 0.000 pixel under the law as built (FAIL), the $\gamma = 0$ ring inside its four pins, and, on the split ladder, the deciding $\gamma = 1$ ring read at 0.731 against 0.731 ± 0.025 with every arrival Node the map's (DETECTOR), the constant through the stated factors outside the one-grain pin at the three $\gamma = 1$ rings by 2.40, 1.07, 1.40 grains (COMPUTATION): the ring's rule read, the bending's 4 not closing within the grain. The run after the algebra is Table S1, row 13.

4.4 Measurement and the quadratic form

The worlds of this section have no body: the law of Section 5.1 on rows alone, and the measurement is the one threshold that is read out. Every interval applies three of the six operations to the state, the translation (the flight, the phase, every count and the birth), the multiplication by a declared integer matrix (the split and the rotation) and the sum in the group ring (the merge, where opposite phases cancel); between clicks the rates are constant, so the map has a closed form and every count of Section 7.1 was computed from it before its run. The click is the one place with a fourth

operation: the apparatus sums a record's rows through the tables and squares the sum, the norm, and only then applies its threshold to the birth phase u .

The row and the record, the update rules of one interval and the click are Definitions 1, 2 and 3 of Section 2, with Lemma 1.

The two theorems of the interval. Two theorems of the linear block are stated here and proved in the Supplementary Material, S5.

Theorem 4 (The split is an isometry whose conjugate transpose inverts it). *The split $(w, \mathbf{m}, p) \mapsto ((wa_i, \mathbf{m}A, p + t_i))_i$ preserves $\sum w^2/\mathbf{m}$ for every nonzero integer vector (a_i) , and the conjugate transpose of the table (the same weights with the turns reversed) applied to its outputs, followed by the normal form, returns the input up to the scaling $(kw, k^2\mathbf{m})$ with $k = A$. The rotation satisfies $U_s^\top U_s = n_s I$ exactly with $n_s = C'[s]^2 + S'[s]^2$, and multiplies a record's norm by $n_s/65536$: an isometry only up to the tables' rounding (65705/65536 at $N_\phi = 64$, $s = 1$).*

Theorem 5 (The interval is injective between clicks). *Let Q be the quotient of $\bigoplus_r M_r$ that forgets the age (rows differing only in their age identified), and fix the event history of the record, the intervals and Nodes of its births, splits, rotations and re-emissions, which the record keeps. For an interval without a click, an end at a face or the border, or a collision, and for splitters whose joint table B (one column per input, the entries $a_i x^{t_i}$ in the group ring) satisfies $B^H B = AI$, the balanced splitter with the turn $N_\phi/4$ among them, $M \circ R \circ S \circ F$ induces an injective $\mathbb{Z}$ -linear map of Q into itself: no rule of the interval deletes a weight, a multiplicity or a phase, and the state at any interval between two clicks, with the history, determines them at the birth. The age is not a coordinate of Q but a function of the history, the current interval less the interval of the row's last event; a split restarts it at 0 by the definition of age (Definition 2) and forgets nothing, since the interval of the split is in the history; the click is the one deletion. A collision, where it acts, is a bijection of the GameBoard's states that is not linear; none acts in the worlds of this paper. The theorem is of the rows' linear block; a body's drive, whose later coincident fire is lost under the engine's rule (P5, Section 5.1), is outside it and is not injective in that case.*

So a record's norm is conserved exactly by S and by the flight, and by R and by every read through the tables only within the tables' rounding (Appendix A). The normalisation by C_K in (4) makes the click's probabilities sum to one at completion; it does not restore the norm along the way, and the paper claims exact unitarity nowhere.

The click's square as a positive quadratic form: a quadratic read-out characterization.

The row of Table S5 on the click's weight is written out here from the derivation's section 6.5 [22]; the register's checks are series L's clicks (Section 7.1). The click's square, the norm of a record's summed pointer, is the one operation of the law beyond the translation, the wall, the declared matrix and the addition, and the place where the Born rule enters. Asked: whether the operations of Section 5.1, with one hypothesis on the reading, force the reading at a click to be that square, or what weaker statement holds and what the register pins.

The objects are those of Section 2: the record's element f of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its pointer $\text{ev}(f)$ and the function R of a cell's element.

The hypotheses. (a) The phase rotation, $R(xf) = R(f)$: a rotation of the record's phase by one step changes no weight (the rungs take no u ; u only selects among them), an operation of Section 5.1. (b) The balanced splitter conserves the reading for every two inputs, $R(x^{N_\phi/4}f + g) + R(f + x^{N_\phi/4}g) = A_2(R(f) + R(g))$ for all f, g (the (1, 1) split with the quarter turn on the reflected row, A_2 its

multiplicity factor): a hypothesis on the reading, stronger than Section 5.1's split, which is an integer matrix on rows and says nothing of what a detector reads under it; the register checks it only on the registered pairs (the totals 64/0 and 32/32 of Section 7.1). (c) Counts, $R(f) \geq 0$. (d) The empty reads zero: $R(0) = 0$ on the zero count vector. (e) Some input reads: R is not identically zero. Without (d) the hypotheses admit a detector that always reads 1, which satisfies (a), (b) with $A_2 = 1$ and (c) and is no quadratic form; without (e) they admit the zero detector, under which (b) holds for every A_2 ; so (a) to (c) alone do not force the square. (d) and (e) are what the engine does (the Gram form on the empty record is 0; a click needs a live unit) and what the proof's first step uses ([6, record 340]; the derivation's 6.5 restated with them). Not assumed: continuity, a dimension, the form of R on one row, or Theorem 4's $A = \sum a_i^2$.

Theorem 6 (The quadratic read-out on the phase lattice, a Gleason-like characterization). *Under (a) to (e), with N_ϕ a power of two, $A_2 = 2$ and*

$$R(f) = \sum_{j \text{ odd}, 1 \leq j < N_\phi/2} c_j |\sigma_j(f)|^2, \quad c_j \geq 0, \quad \sigma_j(f) = \sum_p f_p e^{2\pi i j p / N_\phi}, \quad (12)$$

the σ_j the Galois conjugates of ev , $\sigma_1 = \text{ev}$. Every such R is a positive quadratic form on the lattice, homogeneous, $R(af) = a^2 R(f)$; it conserves the total at every declared split of one row or of two recombining inputs exactly if and only if the multiplicity factor is $A = \sum a_i^2$, and at every label rotation with the factor $C'^2 + S'^2$, C' and S' the half-angle tables of Definition 2. Two rows ax^p and $bx^{p+\Delta}$ at one Node, Δ their phase difference in steps, read $R = C_\Sigma(a^2 + b^2) + 2ab\mathcal{K}(\Delta)$ with $C_\Sigma = \sum c_j$ and $\mathcal{K}(\Delta) = \sum c_j \cos(2\pi j \Delta / N_\phi)$, $\mathcal{K}(0) = C_\Sigma$, $\mathcal{K}(N_\phi/4) = 0$, $\mathcal{K}(N_\phi/2) = -C_\Sigma$ for every admissible R ; the Born form $|\text{ev}(f)|^2$ is $c_1 = 1$ and the other $c_j = 0$.

Proof. Sketch; the derivation's 6.5 [22] gives the steps in full under (a) to (e). (i) $g = 0$ in (b) with (a) gives $2R(f) = A_2(R(f) + R(0)) = A_2 R(f)$ by (d), so $A_2 = 2$, one f with $R(f) > 0$ sufficing, and (e) supplies it. (ii) Replacing g by $x^{-N_\phi/4}g$ in (b) and using (a) and $x^{-N_\phi/2} = -1$ gives $R(f - g) + R(f + g) = 2R(f) + 2R(g)$, the parallelogram law on the abelian group $\mathbb{Z}[\zeta_{N_\phi}]$. (iii) The parallelogram law alone makes R a quadratic form: on a lattice the argument of Jordan and von Neumann [77] needs no continuity, and $R(f) = B(f, f)/2$ with B a $\mathbb{Z}$ -bilinear symmetric form, the restriction of the real form of a symmetric matrix $\mathbf{G}$, the click's Gram matrix (not the fan's grain N_θ of Section 5.1). (iv) By (a), $\mathbf{G}$ commutes with the signed permutation of the basis that multiplication by x is, whose eigen-functionals are the σ_j , j odd; so $\mathbf{G}$ is diagonal in that basis with real entries paired under $j \leftrightarrow -j$. (v) A negative c_j would make the form negative on an open cone containing lattice points, against (c). (vi) The conservation at a general split and at a label rotation follows from homogeneity, (a) and the cancellation of the cross terms between the two outputs. (vii) Two rows: $|\sigma_j(a + bx^\Delta)|^2 = a^2 + b^2 + 2ab \cos(2\pi j \Delta / N_\phi)$. $\square$

What is derived and what is not. Derived from the update rules and the axiom (b): the reading is a positive quadratic form of the record's element, forced by the rotation, the balanced splitter's conservation and the counts alone, with no continuity and no dimension (Gleason's theorem [70] needs dimension three; the lattice version does not, the parallelogram law being granted by the balanced splitter); the multiplicity rule $A = \sum a_i^2$ and the square as the only power are derived, and the cross term $2ab\mathcal{K}(\Delta)$ is forced at the click. Not reached by the algebra: the constants c_j , the detector's response to the harmonics of the phase; a single harmonic j is the Born form under the relabelling $p \rightarrow jp$, a mixture a different law, the flat mixture the phase-blind detector, which interference excludes (Table S1, row 2a; the built click). As N_ϕ grows the admissible readings are the same family, the cone of the even positive-definite functions on the circle with odd harmonics only; no N_ϕ and no continuity forces $c_j = [j = 1]$, and the fundamental is what the register reads.

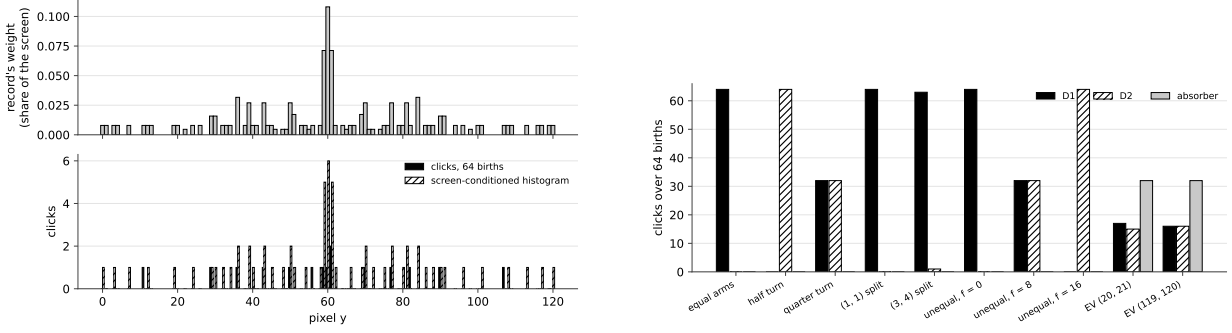


Figure 4: Interference read at a detector, from the runs of series L at $N_\phi = 64$ (the figures' runs, the Supplementary Material, S3). Left, the two-slit world `slits_huygens`: above, the record's weight per pixel of the screen, a number of the board (GAMEBOARD, a diagnostic); below, the clicks over 64 births (DETECTOR) and the histogram conditioned on the screen, bright where the two rows' ages differ by whole turns. Right, the Mach-Zehnder worlds: the clicks over 64 births at the two ports and the absorber (DETECTOR), equal arms 64/0, a half turn 0/64, a quarter turn 32/32, the (1, 1) and (3, 4) splits, the unequal arms and the two Elitzur-Vaidman worlds (Table S1, row 2b; the Supplementary Material, S3).

Against the registered integers. The built click carries the square: the (3, 4) split's 63/1, 31/1 and 125/3 at $N_\phi = 64, 32$ and 128 [6, record 363] hold if and only if the power k lies in $[1.917, 2.012]$, containing 2 and excluding 1 and 3, a check that the implementation is the form of Theorem 6 and no evidence about nature's power (Table S1, row 2c); the norm range of Appendix A bounds one phasor's error between the ideal reading and the tables; the built weight is within $A\sqrt{2R^*} + A^2/2$ of the exact weight R^* for a record of total amount A (a part in 181 at full scale, the one-row extreme a part in 276), with no relative bound where the record's rows cancel below $A/2$ per component: $256x^8 - 181(1 + x^{16})$ at $N_\phi = 64$ reads 0, 196 and 25028 at three rotations against the exact 49 ([34], a computation from the tables, not a run). The two-slit period 23.5 pixels and Malus at 22.5 degrees (219/256 exact) admit the harmonic $j = \pm 1 \bmod 8$ and exclude every mixture of weight above $w^* = 1/1516 = 0.00066$ on $j = \pm 3 \bmod 8$ (w the part of $\sum c_j = 1$ off the fundamental's class, read as the count of 256 that falls from 219 to 218), and the two-slit bands a weight above about 0.02 on $j = \pm 1 \bmod 8$ (one click at the peak of 51); below these weights the register does not exclude a mixture, Malus at 22.5, 45 and 90 degrees reading every member of $j = \pm 1 \bmod 8$ alike ([34], a computation): to these weights the register pins the fundamental, and the least-rank click (P10) stores it.

Interference, by the formulas alone. Every phenomenon of this paper is read the same way: Outside, what a detector counts; down to Inside, the rows, their phases, the merge and the evaluation; back Outside, the click's counts over the births against the reading. Two slits: one record is born with two rows of amount 1, one per opening, $m = 2$; each row's phase advances one turn per Link; at pixel x the two arrive with phases $p_1(x), p_2(x)$, their difference $\Delta(x)$ the path difference times the turn; the merge adds them and an equal pair with $\Delta = N_\phi/2$ cancels to no row; the click evaluates the sum and squares it, the offer $R(x) = 512 [(C[p_1] + C[p_2])^2 + (S[p_1] + S[p_2])^2] = 1024 (65536 + C[p_1]C[p_2] + S[p_1]S[p_2])$ up to the tables' rounding, proportional to $1 + \cos \Delta(x)$; the rungs b_x of Eq. (4) lay the pixels on the ladder, the click is the pixel under which u falls, and over N_ϕ births the count at x is $b_x - b_{x-1}$, Born's to $1/N_\phi$, with the exact tables. With the declared tables at $1/256$ the weights vary with the birth phase within the tables' rounding (hypothesis (a)

of Theorem 6 holds only to that rounding), so the count over the births departs from the rung difference by up to that rounding: two offers of equal ideal weight at the phases u and $u + 1824$, $N_\phi = 8192$, read 4104 and 4088 for 4096 each, a departure of $8/N_\phi$ (a computation from Eqs. (5) and (4), not a run); the built click's bound is the tables' rounding, the theorem's $1/N_\phi$ its ideal. The spacing is the law's own: $\lambda = cN_\phi d/n$ Links, one turn of the phase per age per fringe, Young's $\lambda D/s$ only in the paraxial limit (rung 2, the fan) [23]; the bands' centres are 23.5 pixels apart for the exact two-path law's 23.3 (L2b, DETECTOR). What passes is a pixel with Δ near 0, a wide rung; what does not is a pixel with $\Delta = N_\phi/2$, the rows cancelled, a rung of width 0 where no click can land; one slit is one row per pixel, nothing cancels, a flat count within the fan's grain and no dark pixel; the Mach-Zehnder's two ports are the same algebra at two Nodes, the bright port 64 of 64 clicks and the dark port 0 (row 2b; the Supplementary Material, S3).

Planck and de Broglie, as identities. Under the declared dictionary the release's cost rule reads as $E = h_q s = (h_q N_\phi) f$ and the turn rule as $\lambda = h_A/p$: Planck's and de Broglie's relations are identities of the update rules with one constant, $h = h_q N_\phi = h_A$, an input (Table S5); no detector reading of them is registered.

The uncertainty relation. On the identification of the circle as position and its transform as momentum, a hypothesis of the program and not a derivation ([22], section 22), the support bound $|\text{supp } f| |\text{supp } \hat{f}| \geq N_\phi$ [71], the entropic bound [76] and Kennard's form $\Delta x \Delta p \geq \hbar/2$ in the limit [72, 75] are the transform's mathematics on $\mathbb{Z}_{N_\phi}$ and nothing of the law; the entropy identity of the click, bits read plus bits not read of u equal to $\log_2 N_\phi$ per record, is an identity of the wheel. What makes it an uncertainty is the one read-out: a record read at a Node gives its position and loses its phase, read on the fan's angle it gives its momentum's direction and not its Node; the same one read-out is what denies Bell a hidden joint distribution ("The one fact and its two faces", Section 3; [48, section 10]).

4.5 Bell and CHSH: from weights to counts

The correlation at the settings (a, b) is $E = (c_{++} + c_{--} - c_{+-} - c_{-+})/N_\phi$ over N_ϕ births, one per u , and the CHSH sum is $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$ at the labels $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$. Two theorems follow from the definitions, exact on the GameBoard.

Theorem 7 (Exact count marginals). *For a pair with labels $\{0, 1\}$ of equal weight and settings a, b , let $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$ and $R(o_A, o_B) = J^2$. Then the first party's outcome is + exactly for $u < N_\phi/2$ and - for $u \geq N_\phi/2$, for every (a, b) and every N_ϕ ; and the second party's count of + is exactly $N_\phi/2$ for every (a, b) unless $2N_\phi R(+, +)/C_K$ is an odd integer (a tie of the rung), in which case it is $N_\phi/2 + 1$. No tie occurs for any setting pair at the twelve grains from 8 to 1024 of [5], nor at the CHSH labels for any multiple of 8 up to 4096.*

Proof. Write $n_a = C'[a]^2 + S'[a]^2$. Since the rows of U_b are orthogonal on the integers, $\sum_{o_B} J(o_A, o_B)^2 = n_a n_b$ for each o_A , so $C_K = 2n_a n_b$ and the coarse cumulative after the first party's + is $C_K/2$: $b = \lfloor (N_\phi C_K + C_K)/(2C_K) \rfloor = \lfloor (N_\phi + 1)/2 \rfloor = N_\phi/2$. For the second party, $R(o_A, o_B) = R(-o_A, -o_B)$ (the rows of U_s are each other's quarter turn), so with $x = N_\phi R(+, +)/C_K$ the counts are $c_{++} = \lfloor x + \frac{1}{2} \rfloor$ and $c_{-+} = b_3 - N_\phi/2$ with $b_3 = \lfloor (2N_\phi(C_K - R(+, +)) + C_K)/(2C_K) \rfloor = N_\phi + \lfloor \frac{1}{2} - x \rfloor$. Hence $c_{++} + c_{-+} = N_\phi/2 + \lfloor y \rfloor + \lfloor 1 - y \rfloor$ with $y = x + \frac{1}{2}$, which is $N_\phi/2$ when $y \notin \mathbb{Z}$ and $N_\phi/2 + 1$ when $y \in \mathbb{Z}$. The tie condition is $2x \in 2\mathbb{Z} + 1$, that is $2N_\phi R(+, +)/C_K$ odd. The absence of ties at the listed N_ϕ is the computation over every setting pair [5]. $\square$

No-signalling in the arrival intervals and in the counts: the arrival interval at either arm is that arm's flight alone, independent of the other arm's setting and length (Definition 3, the flight blind) (that interval the host's tick at the arm's set, GAMEBOARD [6, record 768]; the arm's own count is read only under the later key `clock_stamp`, outside this paper's law); neither party's recorded outcome exists as a row before the completion, whose interval is the last arm's, so the availability of the record, unlike the arrival, waits on the far arm's length, and the paper claims no independence for that latency; the first party's outcome is determined by u at the record's birth (Theorem 7) and the second's at the completion by the joint weights, both written at the completion, with the marginal $N_\phi/2$ for every setting (Theorem 7); so neither the counts nor the arrival times at either arm carry the other arm's setting or length, and the one non-local step of the law (P6) is the assignment of the second party's outcome from the joint weights at the completion, on the labels; its interval, the record's, is the last arm's. What the theorem does not cover, stated here. The wheel of the registered pair worlds runs at the rate 1 against N_ϕ , one birth per u in order, so the first party's outcomes over the births are $N_\phi/2$ of $+$ followed by $N_\phi/2$ of $-$, a square wave whose serial correlation over a cycle is $1 - 4/N_\phi$ (0.9990 at 4096), a prediction of the sequence in time that no run reads and no row compares with nature's record of outcomes, which shows no such order. The marginal is for settings fixed or chosen independently of the wheel: a setting chosen from u ($a = 0$ for $u < N_\phi/2$ and $a = N_\phi/2$ after, $b = 0$) gives the second party $+$ at every one of the 64 births at $N_\phi = 64$. On the register the settings are declared per world or balanced over every u (the choosers' 960 births meet every u with every pair once), so nothing chooses a setting from u ; nature's freedom-of-choice loophole has its counterpart here in the wheel, not closed by the register, and the reading that would test it is a marginal off $N_\phi/2$ with the settings chosen from u [45]. And the order of the second party's outcomes carries the first party's settings even when they are independent of the wheel: the cells give $B = A$ at $a = b$ and $B = -A$ at $a = b + N_\phi/2$ exactly, so with $b = 0$ a first party keeping $a = 0$ leaves the second a serial correlation $\langle B_t B_{t+1} \rangle = 15/16$ over 192 births at $N_\phi = 64$, and one cycling $a = N_\phi/2, 0, 0$ (the period 3, coprime to N_ϕ , the independence criterion above) leaves $-5/16$, the marginal 96/96 in both, the same at every shift of the wheel and of the chooser (a computation from the theorem's cells, not a run): the second party reads one bit of the first's choice from the order of its own outcomes. So the theorem's no-signalling is of the counts over a cycle and of the arrival times; the sequence in time signals under the sequential wheel, a result of the law as built and a FAIL of any claim beyond the counts, open until the wheel is other than a counter, which P4 (no draw) does not allow, or the claim is kept to the counts, as it is here. Stated plainly: the law as built, with its sequential wheel, does not meet operational no-signalling; its counts are blind to the other party's setting, the order of its outcomes is not, and until the wheel is other than a counter the pair's construction models a Bell experiment in its counts only, not in its record of outcomes. Read after a detector on the registered pair world (Table S1, row 1c): the second party's serial correlation 15/16 under a constant setting and $-1/16$ under the period-3 cycle 0, 21, 42 (the cycle $N_\phi/2, 0, 0$ the register's keys cannot declare; the declarable cycle reads the same channel with a smaller swing), the marginals 96/96 under both, the algebra met at 16 of 16 checks, nature's value 0: FAIL [31].

The strict-crossing rung $N_\phi C_k > u C_K$ would give the second party 33/64 in 3944 of the 4096 setting pairs at $N_\phi = 64$, a signalling of $1/N_\phi$ made by the rounding [3]; the nearest rung is what the theorem needs. The theorem's scope is what it states: a pair of two labels with equal weights, the CHSH form, and the rung of (4); unequal weights and more than two labels are not covered, and no-signalling for them is open. That the marginal at B is $N_\phi/2$ for every a although B 's rung is computed from $R(o_A, o_B)$ with a and o_A supplied by the layer (Definition 3) is the content of the theorem; the joint counts are not local, and the model does not claim they are.

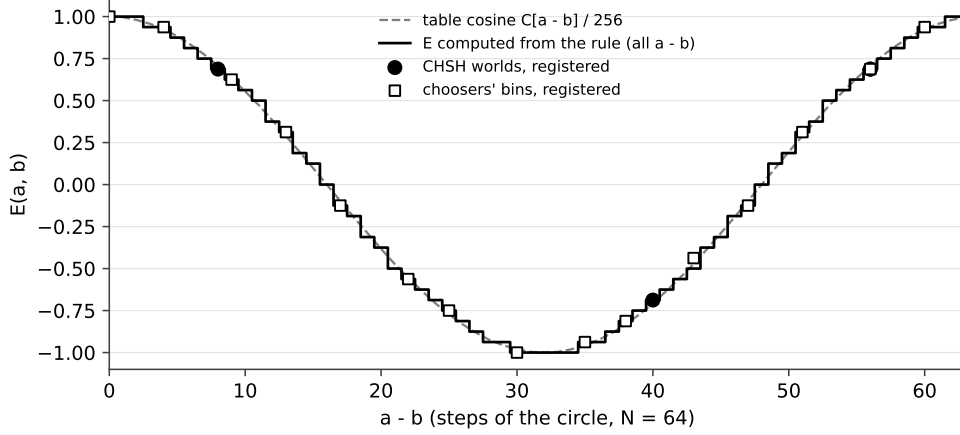


Figure 5: The pair at $N_\phi = 64$: $E(a, b)$ computed from the rule at every setting difference $a - b$ (the counts of Theorem 7 with the repository's tables, a computation [5]), the tables' cosine $C[a - b]/256$ dashed, and after a detector the registered CHSH worlds (circles) and the choosers' bins (squares) of series L (DETECTOR).

Theorem 8 (The finite- N_ϕ Bell value). *With the counts of Theorem 7 and off a tie, $E_{N_\phi}(a, b) = 4c_{++}/N_\phi - 1$ and*

$$|E_{N_\phi}(a, b) - \cos(2\pi(a - b)/N_\phi)| \leq \frac{2}{N_\phi} + 4 \arcsin \frac{\sqrt{2}}{2\varrho} < \frac{2}{N_\phi} + 0.0111, \quad \varrho = N_t - \frac{\sqrt{2}}{2}, \quad (13)$$

so at the labels $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$, $|S(N_\phi) - 2\sqrt{2}| \leq 8/N_\phi + 0.0444$. $S(N_\phi)$ is an exact rational for each N_ϕ ; it lies above $2\sqrt{2}$ for 252 of the 512 multiples of 8 up to 4096 (among them $N_\phi = 16, 32$ with $S = 3$ and $N_\phi = 128$ with $S = 23/8$) and equals $181/64 = 2.828125$ at every power of two from 512 through 8192 and $5793/2048 = 2.828613$ at 16384 [5]; measured after a detector at 512, 1024, 2048, 4096, 8192 and 16384 [4] (Figure 6).

Proof. $c_{++} = \lfloor x + \frac{1}{2} \rfloor$ with $x = N_\phi R(+,+)/C_K$ gives $|c_{++} - x| \leq \frac{1}{2}$ and $E_{N_\phi} = 4c_{++}/N_\phi - 1$ from the mirror symmetry, so $|E_{N_\phi} - (4R(+,+)/C_K - 1)| \leq 2/N_\phi$. The vector $(C'[s], S'[s])$ differs from $256(\cos \theta_s, \sin \theta_s)$, $\theta_s = \pi s/N_\phi$, by at most $\frac{1}{2}$ per component, so its direction differs from θ_s by at most $\delta = \arcsin(\sqrt{2}/(2\varrho))$ with $\varrho = N_t - \sqrt{2}/2 = 255.29$, below every table vector's length (each component within $\frac{1}{2}$ of $256 \cos$ or $256 \sin$), so $4\delta = 0.01108 < 0.0111$ and $16\delta = 0.04432 < 0.0444$; $J(+,+)/\sqrt{n_a n_b}$ is the cosine of the angle between the two directions, $\cos(\theta_a - \theta_b + \epsilon)$ with $|\epsilon| \leq 2\delta$, and $4R(+,+)/C_K - 1 = 2 \cos^2(\cdot) - 1 = \cos(2(\theta_a - \theta_b) + 2\epsilon)$, within 4δ of $\cos(2\pi(a - b)/N_\phi)$. The four settings add the bounds. The values of $S(N_\phi)$ are the computation from these formulas with the repository's tables [5], which reproduces the design's 176/64, 720/256 and 2896/1024 at $N_\phi = 64, 256, 1024$ and the engine's three values below. $\square$

What Theorem 8 does not say: that $S(N_\phi) \leq 2\sqrt{2}$, false for about half the N_ϕ ; nor that the deficit is $O(1/N_\phi)$ from below. The tables' term in (13) is the resolution $1/256$ of the circle's tables and does not shrink with N_ϕ ; at the CHSH labels the tables' entries are the same at every N_ϕ , so the rounding gives $181/64$ at the powers of two from 512 through 8192 and $5793/2048 = 2.828613$ at 16384 and 32768 (measured at 16384, DETECTOR), and beyond the tables' bound $2N_\phi \leq 65536$ the rungs on the fixed correlations are the closed form $S(N_\phi) = 8(c_1 + c'_1)/N_\phi - 4$ with c_1, c'_1 the

first cell's counts, whose limit with the tables fixed is $186034/65773 = 2.828425$, 2.1×10^{-6} below $2\sqrt{2}$; $2\sqrt{2}$ is the joint limit in N_ϕ and N_t [5, 4]. At a tie $c_{--} = c_{++} - 1$ and E_{N_ϕ} drops by $2/N_\phi$; no tie occurs at the CHSH labels up to 4096.

The finite-grain value, and what would make it a prediction. The construction's number is

$$S_{U24} = \frac{181}{64} = 2.828125 \text{ exactly,} \quad \Delta S = S_{U24} - 2\sqrt{2} = -3.02 \times 10^{-4}, \quad (14)$$

at every power of two from 512 through 8192, with the tables at $N_t = 256$, the wheel at $N_u = N_\phi$, the settings $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$ and a pair of two labels with equal weights; the marginals exactly $1/2$; the correlations at the two pairs of settings 725/1024 and 723/1024 at $N_\phi = 4096$, above and below $1/\sqrt{2}$; 5793/2048 at 16384 and 32768. It was computed in advance and measured after a detector at $N_\phi = 512, 2048, 4096, 8192$ and, the plateau's end, 16384 (the last three [4, at b34fe114]; Figure 6), under the click and the wheel with every pin met (the register's block of the derivation's 24.4); 512 and 4096 were re-run by the second runner [8]. The criterion: the law at a grain is rejected when a measured S differs from $S(N_\phi)$ by more than five standard errors, under the identification of the experiment's settings and detection with the labels and the ideal read-out, which the model does not model; the comparison with nature is Section 7.1. It is a prediction of the law only once the pair's wheel meets no-signalling in the order of the outcomes (above), which the sequential wheel as built does not; until then it is the exact consequence of the declared construction at its grain.

5 The GameBoard: the way the algebra is computed with

The GameBoard is the way the algebra is computed with, not the object: a system of information transfer with costs, read in two ways, the detector's click a measurement and the host's view a diagnostic. This section states the law as one map, its state and its components, its read-out and its code, and the ledger of what the algebra shows on it.

5.1 The law and its implementation

The GameBoard as a system of information transfer. The GameBoard is a system of information transfer, and its only objects are messages, the rows: a row is a tuple (Node, direction, age, phase, emitter, content per unit, record, label, amount, multiplicity), and a Link is a channel that carries at most one step of a row per interval in each direction. Nothing else exists Inside: no field at a Node, no register, no memory beyond the rows present. Each interval does six things to messages and nothing else: it moves a row one Link along its digital line (the translation), turns its phase (the translation on the circle), splits it at a splitter into rows with integer weights (the multiplication by a declared table), rotates a labelled pair (an integer matrix), merges the rows that meet at a Node with equal words, opposite phases cancelling (the addition), and, at a detector, reads a record once by one comparison and deletes it (the evaluation and the count). The books say that no message is lost or made in transit: released equals in transit plus absorbed plus escaped plus cancelled at every interval. Two messages of this paper: a row on the heading $+x$ crosses its Links at the intervals 1, 3, 5, 7, 8 (the hand-worked update below), nothing kept at a Node; and at a balanced splitter one row of amount 1 becomes two of amount 1 and multiplicity 2, which meet again at a second splitter, at one port with equal phases (the words add) and at the other $N_\phi/2$ apart (the message cancels), so one detector reads the record, 64 of 64 births, and

the dark port is the cancelled message (series L, Section 7.1). Written down, a message is a basis element of a free $\mathbb{Z}$ -module, the passage is the translation group's shift, the split and the rotation are integer matrices, the merge is the group ring's addition and the read-out is one bilinear form: the passage from physics to modern algebra is the passage from messages with attributes to vectors with coordinates. The lattice gases [53, 54] carry bits on links with collision tables, the cellular automata machines of [30] and the quantum cellular automata of [80] are local rules on cells, and Shannon's channel [7] is the message and its capacity; the GameBoard is of that family, with two differences: its messages carry a phase on a bounded circle and an integer amount, and its only read-out is the click.

Rules, parameters, initial conditions. The law is stated as rules, apart from the parameters and the initial conditions a world supplies. The rules: (P1) every quantity of the state is a bounded integer and every operation integer arithmetic, no real number, root or floating point at run time; (P2) space is a cubic array of Nodes, each joined to six neighbours by Links through directional Ports, a Node's complete local information its NodeState, time a sequence of intervals; (P3) the rates and walls of the map are made of the six operations of Section 1 and nothing else; (P4) a rule reads its own record and the arrivals at its Node and its six neighbours, already delivered, and nothing kept at a Node beyond the events there, no register, remainder or draw; (P5) no row or body crosses more than one Link per interval; (P6) a record is read out once, by one comparison, at the click, the one step that deletes, and the click of a pair is one gather of one record from both settings, the law's one non-local operation. The parameters (P7, P8): the grains, the circle N_ϕ , the pace's grain N_l , the sphere of directions, the birth wheel N_u , the fan's grain, the clock's pair, and the roundings declared at load, among them the circle's tables at $1/256$; and the physical inputs, the family table (per family the content, the cost h per phase step, the charge per unit of content, the strong column, the lifetime, the phase rate and the hand; its instance on the tree [9]) and the world's width N_w , which carries what physics calls Newton's constant. The initial conditions: the box and its boundaries, the bodies with their momenta and contents, the lamps, the detectors, the splitters and the gates of the world file. The law of this paper is the runtime before the generic entry of the bending: the commits of `main` that every run of this paper names in the Supplementary Material, S3, on the first-parent line of `main` up to the archived law's head 3a9a7109, the last commit before the entry's merge 5b855791; the generic entry (`main` at 5b855791 since 2026-09-22, the flight in the age wall's set at $1+\gamma$, γ an input [35]) is a later rule of the law, and the paper's only readings under it are the ring's (Table S1, row 13). Four rules are chosen among few (P9) and are assumptions, not derivations: the flight is blind to the crowd, a row's rate in transit constant and its wall the direction's T_D ; a body's drive is the flight at the fraction its momentum earns, $|p_a|/(N_l N_w M + |p_a|)$ per axis, each axis's count advancing every interval, the first axis whose count fires making the step and a later axis's coincident fire lost, its wall subtracted and nothing carried (the axes x before y before z ; the engine's rule [19]), so that a body crosses at most one Link per interval (P5) and the per-axis fraction is its pace only where the fires do not coincide: three equal momenta $N_l N_w M$ would give $1/2$ per axis, $3/2$ in all, and the built body moves on x alone at $1/2$, the sum of a body's paces bounded by 1; the directional drive of the momentum's line, form B [6, record 652], exists as a declared identity off by default (the world key `drive.b`, series Y), no coincident fire lost, read inside its pins on six registered worlds; it becomes the default only by a declared decision of the author, not before; until then a diagonal body is the per-axis walk; the collision's permutation is the cyclic shift; and a clock counts the age moment of the rows at its Node (the potential's form, [6, record 394]) rather than their presence. One axiom of the apparatus (P10): the click stores one pointer per set, the least-rank member of the family Theorem 6 forces, which is Born's form with

the fundamental harmonic; which harmonic is a relabelling of the declared rate, and the harmonic constants are not derived. One assumption of the frame (P11): the GameBoard, Inside, is read only through its emitters and detectors, Outside; every result Outside is a click, a number of the board's state is never a reading, and every comparison with experiment is a comparison of clicks with readings; a measurement is itself an event of the board, an emitter putting a record on it and a detector ending the record at the click, the one step that deletes Inside and the one row that Outside gains. Nothing outside this list is assumed.

The map. Here s is one accumulator of the state, a bounded integer on a circle or a line; r its declared rate, signed, a constant of the world for a message in transit and a reading of what arrives for a body; d its declared wall, a positive integer; a the component's cap, 1 where a cap is declared and $a = \infty$ where none is (the whole part; the primitive's `at_most` = 0); and e the event count: every unit of e is an event (a Link crossed, a phase step taken, a count completed, a birth, a push), and nothing else happens; a reversed rate first cancels what was accumulated the other way. For a count without cap the third step is the division toward zero with its signed remainder kept, the Euclidean division for $s \geq 0$; under the cap the residual above the wall is kept and fires one event at each following interval until spent. The cap $a = 1$ is the drive's alone, one Link per interval; every other count takes the whole part and can fire several events in one interval, the turn, the owed count, the release, the lamp, the push, so that a rate of 5 against a wall of 2 fires 0, 2, 5, 7, 10 events over the intervals, as the high-rate runs read [6, record 340]. The 0/1 form $e \leftarrow [s \geq d]$, one event bit per interval, holds exactly where the rate never exceeds the wall: the flight ($2S_1N_l \leq 2T_D$, at most one Link per interval), the drive by its cap and with the sign ($e \in \{-1, 0, 1\}$), the birth wheel (the rate 1 against N_u) and a clock whose rate is below its wall (the derivation's section 0 [22]). The rates and walls of every component are made of six operations and nothing else [21, 18]:

1. the translation of an accumulator by its rate (the flight, the phase, the age, every count);
2. the multiplication by a declared integer matrix or bilinear form (a split, a rotation; a coupling, the click's Gram matrix);
3. the addition in the group ring $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ (the merge, where opposite phases cancel);
4. the permutation of the joint state (a collision, a gate);
5. the evaluation of the record at the N_ϕ th root of unity, $\mathbb{Z}[\mathbb{Z}_{N_\phi}] \rightarrow \mathbb{Z}[\zeta_{N_\phi}]$ (N_ϕ the phase circle's width), and its norm (the click's one quadratic step);
6. the division toward zero with the remainder kept (the Euclidean division for a non-negative accumulator), and the comparison (the carry that is the event; the cap; the threshold).

Two places are not linear and there is no third: the carry, a bijection $s \mapsto (e, s - ed)$ with $|s - ed| < d$ and the remainder's sign that of s , which loses nothing, and the click's threshold, the one comparison read out per record, whose remainder is not read: the one place the law deletes by design. The cap and the sign are operation 6's comparison, $\min([|s|/d], 1) = [|s| \geq d]$; the capped carry is the injection $s \mapsto (e, s - ed)$, which loses nothing either, so there is still no third place. Written as algebra, the six are: the carry a bijection $s \mapsto (e, s - ed)$; the linear block's closed form $[s_0 + rt]$, the flight the translation group's shift along the digital line; the split's table with $A = \sum a_i^2$ inverted by its conjugate transpose (Theorem 4); the rotation an integer matrix with $U_s^\top U_s = n_s I$; the merge the addition in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its cancel the quotient by $x^{N_\phi/2} + 1$, which for N_ϕ a power of two is the ring of cyclotomic integers $\mathbb{Z}[\zeta_{N_\phi}]$, a free $\mathbb{Z}$ -module of rank $N_\phi/2$; and the click one bilinear form

$\mathbf{f}^\top \mathbf{G} \mathbf{f}$ with $\mathbf{G}$ the tables' Gram matrix, then one comparison. The exact evaluation at the roots of unity and the evaluation by the tables at $1/256$ are two maps: the cancel's zero, an equal pair at $N_\phi/2$, is a zero of both, the click's other zeros are the tables' own (Section 2), and each claim below says which of the two it is about. Modern algebra is not assumed here; it is what the cell rules are when written down, and every theorem below is a property of these six maps.

The state. The GameBoard is a box of Nodes in $\mathbb{Z}^3$, each joined to six neighbours by Links through six Ports; an axis may be periodic. Time is the interval $t \in \mathbb{Z}$. The state at an interval is a vector $\mathbf{s}$ of bounded integers on a product of circles and lines, and it is the multiset of the messages in transit (the rows) and the bodies (the measured events), and nothing else: a Node holds no register, remainder or draw. A body carries its Node, its content M , its momentum vector $\mathbf{p}$ and its table of counts, one accumulator per count (the fields per kind on the tree [9]).

The components of Eq. (1). The ladder's rungs b_k of Section 2 are walls of this kind. The components and their (r, d) [22]: a row's Manhattan count (the rate $2S_1N_l$, the wall $2T_D$, started at T_D ; D the direction vector, S_1 its Manhattan length, $T_D = \lfloor \sqrt{3|D|^2N_l^2} \rfloor$ its resolution, N_l the flight's grain), with three axis deficits choosing the axis of each Link; a row's phase (an integer per Link, or a declared rational ν of steps per interval, written n/d in Section 2, its whole part read modulo N_ϕ); a row's age (the rate 1, no wall; bounded by the family's lifetime and the run); a body's drive on each axis (the rate $|p_a|$, the wall $N_lN_wM + |p_a|$, M the content and N_w the world's width), whose speed $p/(N_lN_wM + p)$ per axis where the axes' fires do not coincide, one Link per interval in all, saturates at one Link per interval, above the rows' pace; a body's counts (the turn, the owed count, the release, the lamp, the push per column, each a rate against a wall); a record's birth phase (the rate 1 per birth, the wall N_ϕ ; the birth wheel, a lamp's declared rate r_w per birth against the wall N_u , [2531, 4096] on the registered two-slit world, the rate the nearest odd integer to 0.618×4096 [2]); the exact phase at the click [2]; and the click's ladder, one comparison read once. Beside the six operations stand the roundings declared at load, each a constant of the law, among them the tables of the circle at $1/256$, the unit vector of a direction to the nearest integer vector, T_D by the integer square root, the click's rung to the nearest integer and the fan's weights at the grain N_θ [18].

The two blocks. Φ is linear where its rates are constants of the world and feeds back where a rate is a function of what arrives. The linear block is the rows in flight: the flight, the phase and the age at constant rates, the merge, the integer matrices of a split, a rotation or a gate, and the evaluation at the click. Every component of it has the closed form of its unwrapped count, $\lfloor s_0 + rt \rfloor$ (the accumulator itself is that count's remainder by the wall), so a row's state at any interval is a formula of its birth and its age, and the limits of Sections 4.2 and 4.3 are limits of that formula. The feedback block is the bodies: the push, whose rate is the reader's content times the flow of the arriving rows (bilinear in the state); the owed count, whose rate is what the body reads (the presence, or the age moment); the turn and the release, whose rate is the content; the drive, whose rate is the momentum the push changes. These have no closed form; they are iterated, and their continuum limit is the differential equation $ds/dt = \mathbf{r}(\mathbf{s})$. The exact word for the whole map is therefore piecewise-linear in the rows with bilinear rates in the bodies [22].

The read-out. The click reads one comparison and nothing else leaves the rows: for a record whose rows have all ended, the apparatus sums the rows' integer phasors per cell through the tables, squares the sums (the one quadratic step, forced in form by Theorem 6), lays the cells on a ladder

and compares the record's coordinate u with the rungs; the cell whose rung u falls under is the click. Between a birth and a click every step of the linear block is injective on the rows' weights, multiplicities and phases for the event history the record keeps (Theorem 5); the click is the one deletion.

A hand-worked update, and the code. One row's update worked by hand, interval by interval, and the engine's realisation of Eq. (1) as one pass per interval are S6 of the Supplementary Material.

The code implements the law. Every registered world is run from its world file with its expectation written before the run and its source fingerprint; a second runner re-ran series L, T and fifteen more worlds bit by bit [8]; the books balance at every interval in every run; the rounding pins of the (3,4) split were met (the Supplementary Material, S3); the crossing rule's tests read their pinned counts. These are gates on the code, not measurements.

When the transition was made. The road from ordinary physics to this algebra was walked in twenty-four dated steps between 2026-09-17 and 2026-09-22, each a thing of physics becoming an object of the law, recorded with the reading that showed it [25]; the Supplementary Material, S3 lists them.

The ledger. Table S5 of the Supplementary Material is the paper's spine: for every result, the rules of Eq. (1) it starts from, the assumptions added beside them with their kind, the freedom left, and the ground of the derivation [21, 2026-09-22]: exact on the GameBoard (no limit, no average), a limit of the grain taken on paper with a rate, or a limit under a spatial condition, an average that covers the shell (over the shell's Nodes, over the directions of a fan declared in a ball, or over the sweep of an orbit), the condition then part of the claim. A result whose derivation is not closed, or whose only reading is of the GameBoard's state, is not in the table; the hypotheses beside the law are on the tree (Section 3).

6 How the algebra was reached: from the GameBoard to the group

The order of this paper is not the order of the finding. The object of Section 2 was reached from the GameBoard: from one Node and its six neighbours, one message per Link per interval, through the simulator's runs, in dated steps; what did not converge to the group and the ring was not put in. This section states the two directions first, as two tables (the postulates and what each forces; the objects and what each gives back, with its word), then the road from the operations to the group, and then the road as it was recorded: the simulator's part, how the things of physics entered the algebra, the dated steps, the platform on which a formula is arrived at by two roads, and the check of the roads.

From the physics to the mathematics: the postulates and what each forces. The road ran from the physics to the algebra: a discrete world and its locality were assumed, and the objects of Section 2 are what those assumptions force. Table 1 has one row per postulate (named in Section 2), the physical assumption on the left and the mathematical object it forces on the right; Table 2 is the way back, what each object gives that is compared with nature, each with its word

(exact; recovered in a limit; reached under named hypotheses; conjectured; compared). The two tables are the paper's two directions.

Table 1: From the physics to the mathematics: one row per postulate, the physical assumption and the mathematical object it forces; the objects are those of Section 2, the theorems those of Section 4.

Postulate	The physical assumption	The mathematical object it forces
P2 the cubic lattice	space a cubic array of Nodes, each with six Ports	the lattice $\mathbb{Z}^3$, the six directions, and their symmetries: the 48 signed permutations, their rotations the group of order 24 (Theorem 1)
P5 one Link per interval; P4 locality	no row or body crosses more than one Link per interval; a rule reads its own record and the six neighbours and keeps nothing at a Node	the translation group's shift; the causal front, the octahedron $ x + y + z \leq 1$ (Figure 3); the Manhattan bound $S_1 N_l \leq T_D$, and the pace $c = 1/\sqrt{3}$ in the limit of the grain (Proposition 1)
P1 bounded integers	every quantity a bounded integer; the phase on a bounded circle	the cyclic group $\mathbb{Z}_{N_\phi}$ and the free $\mathbb{Z}$ -module of the rows; the amount an integer coefficient
P3 the six operations, the merge	messages that meet add, opposite phases cancelling	the group ring $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ and its addition, (G); the cancel the quotient by $x^{N_\phi/2} + 1$
P7 the declared tables; P8 the families as integers	the tables declared at load; every family one row of integers	the integer matrices of the split and the rotation and the coupling matrix, (B); the pair's record in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ with its arms in $\mathbb{Z}^2 \otimes \mathbb{Z}^2$
P3 the six operations, the accumulator	the rates and walls made of the six operations; every count a bounded accumulator against a wall	Eq. (1): the translation, (T), and the division with the remainder kept together with the comparison that is the event, (D)
P6 one read-out; P11 clicks only; P10 the click axiom	a record read out once by one comparison; the GameBoard read in no other way; one axiom of the apparatus	the evaluation at the roots of unity with its norm, (E); the click's weight the positive quadratic form (Theorem 6); the Outside, counts between clicks and their ratios
P9 the blind flight	the flight blind to the phase and to the crowd	no dispersion, the constancy of c ; a rule chosen among few, so named

Table 2: From the mathematics back to the physics: what each object gives that is compared with nature, with its word; the ground of each is a row of Table S5.

The object	What it gives	The word
The books and the walk, the carry a bijection	the conservation and the continuity equation; Gauss's law of a free family's flux under its world condition (Section 4.2)	exact
The six Ports, the group of order 24 and the front	every direction's pace $N_l D /T_D$, isotropic within $1/T_D$; $c = 1/\sqrt{3}$ (Section 4.1)	exact (the pace); recovered in the limit of the grain (c)
$\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, the evaluation and its norm	Born's form at the click and the interference counts; Born's rule; Young's spacing (Section 4.4)	exact (the form, the counts); recovered in a limit (the rule, the spacing)
The pair's record and the one gather	the exact marginals and no-signalling of the counts; $S(N_\phi) = 181/64$ on the plateau; Tsirelson's $2\sqrt{2}$ (Section 4.5)	exact (the rational); recovered in a limit (Tsirelson's value)
The push's bilinear form, averaged over a shell	Newton's inverse square with the place of G ; Coulomb's law; Poisson's equation and the retarded potential (Sections 4.2 and 4.3)	recovered in a limit under a shell average
The step rule, which divides by the content	the equivalence principle; the third law at rest (Section 4.2)	exact; the third law to the apportioning's grain
The clicks under the named hypotheses	under (A1) with (A3) the r -free lines of Lorentz's factors, under (A2) their scale and Einstein's step; under (A1) and five assumptions of the law as built, with no (A2), the equivalence principle and, in a limit under a shell average, Newton's inverse square (Section 3)	reached under named hypotheses
The flow counted per Euclidean Link; the closure congruence of the action rows	the bending's $2(1 + \gamma)$; the atom's $1/j^2$ ladder (Section 7.2)	conjectured from the algebra

The object	What it gives	The word
The detector's counts	the readings of Table S1, each a count between clicks or a ratio of such counts (Section 7)	compared

From the operations to the group. The operations did not deduce a group; they came first, as acts on the board: a row flies one Link per interval along its direction, rows meet at a Node, a body steps by an accumulator against a wall, a phase turns, identical rows merge, a detector clicks; each was then written as one of the six operations on a state vector. The group is the answer to which maps preserve what the operations act on, a Node of six Ports and its Links and a cone of one Link per interval: the bijections of the Ports that keep opposite Ports opposite are the 48 signed permutations, and over the integers they are the only bijections that preserve the cone, a boost being not $\mathbb{Z}$ -linear, so Lorentz's form is reached Outside from the clicks and not as a symmetry of the lattice (Section 3). The 48 first appeared in the code as the invariance of one operation, the collision's table, and were recognised after as the symmetry of all of them; the 24 came from the hand, the pseudoscalar that carrying spin and polarisation needed, which $\det(g)$ keeps or flips. The other objects came the same way: the step and the flight are shifts, hence the translation group; the turn adds modulo N_ϕ , hence $\mathbb{Z}_{N_\phi}$, and the merge of records is the addition of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$; the collision is a permutation of a finite set; and c was not chosen, it is the flight operator's norm. What the group does back is to force forms: a rule that reads only the local state must commute with the 48, and the symmetric integer matrices fixed by the 48 are the multiples of the identity, so the only isotropic even reading of a body's momentum within the six operations is $c\mathbf{p} \cdot \mathbf{p}$, and the square W of Eq. (6) follows; “generic” in the three tests of every rule means exactly this. The definitions and this road, with their code and tests, are the group-structure page of the tree [11]; this paper writes no second definition.

How the simulator led to the results. The simulator ran first and the derivations followed it: every run registered with its expectation written before the run, its world file and its source fingerprint, each registered number traced back to the operations of the map until a derivation stood or the number was marked a finding; a relation defined in advance is a definition or an input. The two roads, the algebra's and the simulator's, are one platform, stated below so that a reader can arrive at a formula the same way.

The road from the cells to the algebra. How the things of physics entered: the amount w of a row is a positive integer, translated and merged; the phase p is an element of $\mathbb{Z}_{N_\phi}$, turned per Link, added in the group ring, evaluated at the roots of unity; the multiplicity $\mathfrak{m}$ is a positive integer, $A = \sum a_i^2$ at a split; the age τ is a count at the rate 1; mass is the content M , a count of units per family and an input of the family table, which sets the drive's wall $N_l N_w M + |p_a|$ (the division), scales the push's bilinear form and the release rate, and is the rest energy in whole units, $E'_0 = E_0/c^2 = N_l N_w M$; charge is ρ , the declared charge per unit of content, entering only the coupling matrix $\rho_A \rho_B$ of the push; momentum is an integer vector, translated by the push, read by the drive as a rate against a wall, its square compared and never rooted; the field a body reads is the label flow $\mathbf{a} = \sum \text{amount} \times \mathbf{u}_D$, bilinear in the arrivals; the family table (content, cost per phase step, charge, the strong column, lifetime, phase rate, hand) is read and never computed; and the couplings, the circle's tables at 1/256, the split and rotation tables and the click's Gram matrix, are integer matrices and one bilinear form.

The transition from ordinary physics to modern algebra, dated. The twenty-four dated steps, each a thing of physics becoming an object of the law with the reading that showed it, are [25] (docs/designs/algebra_transition/HISTORY.md on main at 412f5c61), their dates once: 2026-09-17 the shared resource deleted and the amplitude a phase with a content; 09-18 the Born table from N_ϕ and a Node its six Ports; 09-19 the quantum an integer row, the click the one border, the push one bilinear form, $E = hf$ a release's cost; 09-20 the readings by kind, the masses the initialisation, every count an accumulator, the fan a width of the law; 09-21 the six verbs, the record in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, the lattice the translation group, the Lorentz factor refused as a seventh verb; 09-22 the detector's clock, the click theorem and the framing.

The platform: how a formula is arrived at, by the algebra and by the simulator. Every formula of this paper is arrived at on one platform, by two roads that meet at the detector. The algebraic road has three steps. The Inside step: the six verbs on the record's attributes, the family's declared integers naming which attributes the step reads (Section 5.1; Table S6). The conversion: an emitter puts a record on the board and a detector ends it at a click, so that Outside there are counts and Nodes only, the detector's own count the age wall's member at coefficient 1 (P11). The Outside form: a ratio or a difference of those counts, the image of the Inside step, with its rung (exact, in the fan's limit, in the shell mean) and its order in the grain; a known form, Einstein's or Young's, stands on the comparison side only. The simulator's road runs the same three steps: a world file declares the families, the emitters and the detectors, and the reading is pinned before the run. Every reading of the simulator's road is DETECTOR in its count (each series with its detector in Table S4), the self-creations' ticks of series S GAMEBOARD; where a count is taken at an open-face detector without a body (the redshift rows 3, 4b and 12) it is read at head in the detector's own clock, its count of self-creations under the age wall ($r = 1.0000$, no crowd at the detector; [45], the re-run of 2026-09-22), equal to the host's interval to the integer, DETECTOR; the registered run's time base was the host's interval [6, record 768], superseded; a two-way light clock at the detector, not read, reads the same count in no crowd (series X); and a number of the board's state is a diagnostic, never a reading. A formula enters the paper with both roads: the algebra's chain with its inputs and its word (Table S2), the simulator's reading with its series and kind (Table S4 and Figure 6), and nature on the comparison side (Table S1). The platform is the paper's offer beyond its list: to arrive at a further formula, declare the family and the attributes it reads, write the Inside step on them, name the detector and what it counts, take the ratio Outside, pin it before the run, and read it by kind; the three chains below do this in full.

The road to each, and the check that nothing was derived from them. The three roads are the one chain above, each from the Inside step, the six verbs, (A1) and, where it enters, (A2), with (A3), locality Outside (nothing Outside passes faster than a chain of clicks or jumps a Node), a definition of Outside and not a law [24]: to Lorentz, the boost from two legs at one Node per interval with $k_{AB} = k_{BA}$, the scale r free until (A2) fixes it at $\sqrt{1 - v^2}$ (the click frame's section 2); to Einstein, the rate $1/\gamma$ as r read from two factors and $E^2 = E_0^2 + c^2 p^2$ as the line of the rate with W shown to second order (Theorems 1 and 2 and rows II.1, II.3 and II.6 of [24]); to the step above itself, the Outside step as the image of the Inside step, Einstein's step its most general form where r is supplied, its quantum a theorem of (A1), (A3) and the drive's one Link per self-creation, Newton's step its limit at $v \ll c$ (section 3 of [24]). Einstein's, Lorentz's and the Outside step's formulas appear in these chains only as the thing compared with; no chain takes them as an input, and the Outside step is never a second formula set beside the Inside step. The word is "follows" or "recovered" where a known form is arrived at, "shown to second order" for W under (A1) and (A2),

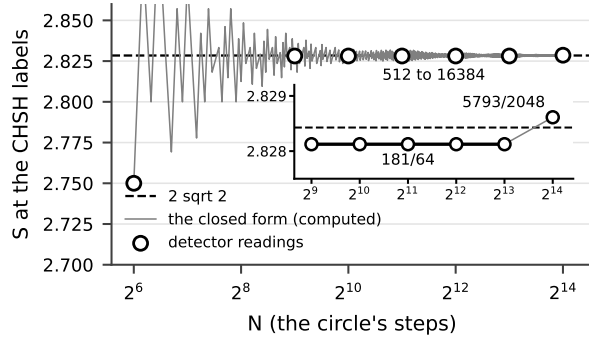


Figure 6: $S(N_\phi)$ at the CHSH labels. The curve is the closed form $S(N_\phi) = 8(c_1 + c'_1)/N_\phi - 4$ computed from the rule at every multiple of 8 with the tables at $N_t = 256$, a computation [5]; the circles are detector readings (DETECTOR) at $N_\phi = 64, 512, 1024, 2048, 4096, 8192$ and 16384 (series L, L6 and the register's block of the derivation's 24.4, the last three [4, at b34fe114]); the inset marks the plateau $181/64$ from 512 through 8192 and its end $5793/2048$ at 16384 ; the dashed line is $2\sqrt{2}$.

“declared” for W on the rows’ dynamics and for the unit $E_0 = mc^2$; “derived” stands nowhere. Table S2 (the Supplementary Material, S2) lists the check row by row, the inputs, the file and line where each chain starts and the word, read against the certification tables of the two sources and audited in their review.

7 The comparison with nature, and the failures by cause

Everything compared here is a detector’s count or a ratio of counts (Section 3); the readings are compared by kind, and each failure is named with its cause.

7.1 The simulator’s checks and the comparison with nature

Three things are kept apart: the gates on the code; what is shown Inside by algebra under named assumptions, which needs no experiment; and what is measured, a run Inside read Outside, compared with known experiments.

The formulas after a detector. Every number read after a detector is a count of clicks or a click’s tick at a declared detector, registered with its source fingerprint (the Supplementary Material, S3; [4, 8]), its expectation written before the run; a number of the GameBoard’s state appears nowhere. Seven formulas shown Inside (the pace, the click’s power, Mach-Zehnder, the marginals, GHZ, Malus, light beside a mass) were confirmed by their runs (the Supplementary Material, S3); the readings that no formula gives are Table S4’s (the clock’s field at two Nodes, the equivalence after a detector, the covariant readings) and Figure 6’s (S at seven grains).

The comparison with nature. The confrontation register [49] keeps one row per registered observable with a dimensionless counterpart in nature, one published source per value; Table 3 gives the comparison in one page, one line per observable; the full record, every row with its numbers by kind, its sources and its notes, is Table S1 in the Supplementary Material, S1: the rows read after a detector and the six that rest on a pin whose run is not made (4a, 5b, 10, 11a to 11c), the law’s own number by a click printed in every row, so that nothing is hidden and no row is removed. The status column takes three values: *agrees*, the reading within the stated tolerance (a bound met is marked so); *disagrees*, with the number, the law’s own result stated with the same care as an agreement; *not predicted* by the law as built, with the missing piece named

in the cell and its kind: (1) the instrument is missing in the model (a moving thing's clock rate, a moving detector's count, a velocity from clicks, the crowd's history), (2) nature's number is a bound or a fitted parameter and not one measurement, (3) no click definition yet for the quantity (a position, a brightness, a period); a row may carry two kinds; the missing piece per row is the one the fail-rows file names [12]. Six rows keep their earlier words (12 history, 2c not compared, 2a NOT COMPARED until its source is verified, 10 not yet, 11b a pin needing a hypothesis not built, 14 not yet read against nature), and 1b is a control, not counted. A claim may stand in two rows, the second marked *beside*, one status each; none is added here. Every number's kind is named (DETECTOR a click; GAMEBOARD the host's view, a diagnostic; COMPUTATION the algebra's); the law's number is the law's as built at the commit of the Supplementary Material, S3, and a reading under a declared hypothesis stands beside it, never in its place.

The limitations, stated before the comparison. The third status follows from the structure of the law as built and never from the outcome, and a referee's test that every cell must pass is that a row sits in the same class had it agreed. The law as built has no relativistic dynamics: a moving thing's own count is $r = 1$ at every speed (Section 3) and nothing contracts, so no observable that reads a moving clock's rate or a moving body's extent is predicted; it has no term of the crowd's history, so no observable that reads the growth of an emitter's stretch with the flight time is predicted; and a declared input is not a prediction, so an observable whose number is a declared give (row 7a), a declared content (row 8c) or a declared coefficient (row 13's γ_{PPN}) is not predicted, whatever the declared world reads. A fan of directions bounded in Manhattan length is not a fan of every direction (its angular density is the octahedron's, the axes 2.8 times denser than the face diagonals); a reading that assumes the isotropic mean holds only where the fan is declared uniform in angle at a grain finer than the reading, or where a click read the crowd; otherwise the law's own number for that world is the integral on the comb (the register's finding under rows 6, 10, 13 and 14). The weight rule, stated once: under the declaration $\gamma_{PPN} > 0$ a moving body's gravitational charge is the pair $w = \gamma_L(1 + \gamma_{PPN}v^2)M$ with its wall, γ_L the Lorentz factor of the body's speed and γ_{PPN} the declared space-curvature coefficient (the parametrized post-Newtonian gamma), declared as that key's rule (the owner's decision B, record 1150 [6]); at $\gamma_{PPN} = 0$ a body weighs M , the law's rule at rest, no velocity term on the body; rows 13 and 14 are read under that declaration where they say so, and the coefficient $2(1 + \gamma_{PPN})$ of row 13 is the declared input and not a result. The count: of the ten observables the law as built predicts (eleven once row 14 is read), four agree within tolerance (1a, 2b, 9; 5a as a bound met), six disagree (1c, 6, 7b, 8a, 8b, 13), and eight are not predicted by construction (3, 4a, 4b, 5b, 7a, 8c, 11a, 11c), each with its missing piece named in its cell and its kind, the instrument missing in the model, nature's number a bound or a fitted parameter, no click definition yet, or the model's own number a declared input; 2a NOT COMPARED until its source is verified; six rows keep their words (12 history, 2c not compared, 2a as said, 10 not yet, 11b a pin, 14 not yet read against nature), and 1b is a control, not counted; the disagreeing rows are the law's own results, and the paper claims no rule it has not shown. Two placements, row 7a to not predicted and row 2a to NOT COMPARED, are the model owner's word of 2026-09-23 after the kind audit [13]; nothing else moved.

The comparison in one page. Table 3 gives one line per observable in the count, with the distance between the law's value and nature's in units of the measurement's standard uncertainty where the source states one; Table 4 lists the rows that keep their earlier words. The notes of every row, the numbers by kind and the readings beside the law's are in the full record, Table S1 of the Supplementary Material, S1, keyed by the same row numbers.

Table 3: The comparison with nature in one page: the observables the law as built predicts, one line each. Nature’s value with its source (“unverified” where the register has not yet checked the figure against the source); the law’s own value with its kind (DETECTOR a count of clicks; COMPUTATION the algebra’s number, no run); the distance in units of the source’s standard uncertainty σ where one is stated; the status in one vocabulary. Notes: ^a a bound or an exact form, no distance defined; ^b the source states no standard uncertainty, so no distance in units of it is given and the disagreement is by the number.

Row	Observable	Nature (source)	The law as built (kind)	Distance	Status
1a	The CHSH sum S of a pair	2.42 ± 0.20 [90]	2.75 at $N_\phi = 64$ (DETECTOR)	1.65σ	agrees
2b	Mach-Zehnder visibility	0.98, apparatus-limited [91], unverified	1.000, 64/0 over 64 births (DETECTOR)	bound met ^a	agrees
9	Malus’s law at 45 degrees	1/2, an exact form	128/256 exactly; 0.8555 against $\cos^2 = 0.8536$ at 22.5 degrees (DETECTOR)	exact ^a	agrees
5a	The anisotropy of c by direction	below 10^{-18} [57], unverified	0.5774 to 0.5818 by direction (COMPUTATION; series Q 290 of 290 clicks, DETECTOR)	bound on N_l^a	agrees
1c	No signalling in the order, the lag-1 serial correlation	0	15/16 and $-1/16$ (DETECTOR)	$-^b$	disagrees
6	Balmer’s line ratio $\nu(H\beta)/\nu(H\alpha)$	27/20 exactly [44]	the loop opens at 3407 intervals, no line read (DETECTOR)	$-^b$	disagrees
7b	The alpha’s binding over the deuteron’s	12.72 [60]	2.0 (DETECTOR)	a factor 6.4^b	disagrees
8a	The neutron’s decay curve, width over median	3.17 [67], unverified	0, a step (DETECTOR)	$-^b$	disagrees
8b	The neutrino’s passage through a second detector	about 1	16 of 1024 and 0 behind (DETECTOR)	$-^b$	disagrees
13	The bending of light at the Sun’s limb	1.75 arcseconds [43], unverified	0.000 pixel (DETECTOR, series K)	$-^b$	disagrees
3	The deceleration parameter q_0	-0.53 ± 0.01 [62]	-0.108 (DETECTOR)	42σ	not predicted
4a	The muon’s lifetime in flight over at rest	29.33 [58], unverified	1 (a body’s own record, DETECTOR)	a factor 29^b	not predicted
4b	A moving lamp’s redshift	0.315 [59], unverified	0.2647 (DETECTOR)	$-^b$	not predicted
5b	The arms’ anisotropy of a laboratory in motion	a null at $\beta = 10^{-4}$ [89]	the round trips γ^2 and γ , a pin (COMPUTATION)	$-^b$	not predicted
7a	The deuteron’s binding fraction	0.1185 percent [60]	0.109 percent, 4 of 3677 (DETECTOR)	$-^b$	not predicted
8c	The heaviest neutrino mass over the electron’s	at least 9.8×10^{-8} [64, 65], unverified	0, a declared content	$-^b$	not predicted
11a	q from a far lamp’s brightness	-0.53 ± 0.01 [62]	+1, a pin (COMPUTATION)	153σ	not predicted
11c	Tolman’s exponent n	4; measured 2.3 to 3.1 [66], unverified	1, a pin (COMPUTATION)	$-^b$	not predicted

Table 4: The rows that keep their earlier words, outside the count; their notes in Table S1.

Row	Observable	Nature (source)	The law as built (kind)	Word
1b	The CHSH sum under the phase-form window	2.42 ± 0.20 [90]	$S = 2$ exactly, Bell’s bound (DETECTOR)	a control, not counted
2a	Two-slit visibility	0.98 a Mach-Zehnder’s [91]; 0.94 a biprism’s [46], unverified	0.966 (DETECTOR)	NOT COMPARED until the source is verified
2c	The Sorkin parameter κ_S	0.0064 $\pm$ 0.0119 [61], unverified	the power’s window [1.917, 2.012] (DETECTOR)	NOT COMPARED
10	The single-opening spread	0.886 [68], unverified	1.08, the crowd form (history)	NOT YET

Row	Observable	Nature (source)	The law as built (kind)	Word
11b	The stretch of a far lamp's stream, $(1+z)^b$	$b = 0.97 \pm 0.10$ [63], unverified	$b = 1$, a pin (COMPUTATION)	a pin, the run not made
12	The clock's field at two distances	2.00 [17], unverified	1.907, the age clock (DETECTOR)	history
14	Newton's periods, the ratio at $r = 24$ and 12	2.00 ± 0.18 , a host-tick period	1.677 and 1.512 (COMPUTATION)	not yet read against nature

The prediction, and the failures. Against the photon-pair value $S = 2.82759 \pm 0.00051$ [88], under four assumptions the paper states and does not defend (nature's pairs this model's pairs at some N_ϕ , fair sampling, the four settings the model's labels, the settings independent of the wheel), the law's 181/64 stands 1.05 standard errors above, 16384 and beyond at 1.5 to 2.0; $N_\phi = 64$ and 256 are excluded; the loophole-free [90] excludes no N_ϕ . The proximity is a compatibility, not an advantage over quantum mechanics; what refutes it is a CHSH measurement at an uncertainty below 1×10^{-4} reading a deficit below 2×10^{-4} or above 4×10^{-4} , and a separation of 181/64 from $2\sqrt{2}$ at five standard deviations needs an uncertainty below 6×10^{-5} before systematic errors. The value is the model's at its declared table scale $N_t = 256$ and at the registered grains 512 to 8192; why nature's grain would lie there, or its table scale at $N_t = 256$, the paper does not say; but the dependence is itself algebraic and not a fit: $S(N_\phi)$ is the exact rational of Theorem 8 at every grain, 181/64 at every power of two from 512 to 8192 and 5793/2048 from 16384 to the tables' bound, so what the model predicts is that family of rationals, computable without a run, and a measurement at the stated uncertainty decides among them or refutes the family as a whole: a value below 2.828125 or above 2.828613 by five standard deviations refutes the registered grains, the powers of two from 512 to 32768, at once; the other multiples of 8 give other rationals of the same theorem, 17/6 at 768 and, over the enumerated grains from 1024 to 4096, from 2.82443 to 2.83212 (the check's scan [5], COMPUTATION, not a bound beyond it), so such a measurement decides among the family's members and refutes the plateau, not every grain; and nature's 2.82759 ± 0.00051 sits 1.05 standard errors below the plateau. The rows that disagree are the law's own results, each with its distance in Table 3; each refutes the law as declared on its own, and no count of agreements weighs against them; the rows not predicted by construction name their missing piece and its kind in their cells, the law's own number printed in each; 1b is a control and not counted, and 14, Newton's periods, is not yet read against nature.

7.2 Discussion and conclusions

The chain from the click to Lorentz's factors, Einstein's step and Newton's form is Section 3; the two roads and their check are Section 6; what is proved opens Section 4.

What the law names and has not computed. Each item below is a computation the law's rules define, on a declared world with a declared click, that this paper has not made: the register holds no reading for it, nothing is claimed for it, and its verdict is NOT YET, NOT COMPARED or absent until a run is made and read by kind. (i) The lines of the atom: not a run not yet made but a rule the law lacks, no rule of the six quantising a body's loop and one click reaching only the loops with $2i^2 \geq j^2$ (Table S1, row 6; [39]); Bohr's 27/20 waits on that rule, under its own identity; the ladder itself, $a_j/a_i = (j/i)^2$ and $E_j \propto 1/j^2$ with 27/20, is conjectured from the algebra, not computed (item (viii) of the list above). (ii) Third-order interference, Sorkin's κ_S on a three-opening world (row 2c; no such world is registered). (iii) The single opening's spread against its pin (row 10), and the rows pinned without their run (4a, 5b, 11). (iv) The quarks and the nucleon: the quark families are in the catalog with their rational charges and their content per

record (Table S6, u), and no detector reading of a quarks world is registered; the nucleon as three measured events bound by the strong column, its read mass against the sum of its parts, is the reading that would test them [6, record 248]; colour, a $\mathbb{Z}_3$ label that passes the three tests and does nothing beside a colour force between rows that fails the vector and the local test, is designed under its own identity and not built [6, record 270]. (v) The unseen content: on the law's own rules the field at a galaxy's edge is Newton's in the shell mean and the detector adds nothing there; the one route to a flat rotation curve without unseen content, a short periodic dimension ([14], entry 6), fails nature's Tully-Fisher slope on a host map ([42], no run); what the law admits is unseen content as an input family, as nature's model does; no rotation curve is read after a detector and no disc under a declared unseen family is run. (vi) The size of the crowd's stretch, $g_1 = 0.36$, that the supernova diagram needs, from a declared content over a flight (an input in the dark energy item of the list above; the crowd's own accumulation gives the wrong sign), and the periodic GameBoard against the open one on the measured numbers, declared and not decided [6, record 270]. (vii) The detector's constants c_j of the click's harmonics, not reached by the algebra (Section 4.4). (viii) The inverse square in space, pending its run (row 14); the bending's full deflection, its coefficient $2(1 + \gamma)$ on the one constant conjectured from the algebra; the ring's rule read inside its pins and the constant 4 not closing within the grain (row 13). Apart from these stand the numbers for which the law has no formula and this paper offers none: the hierarchy ρ_p , the constant $h = h_q N_\phi$, the families' content per unit and the grain N_ϕ , inputs by the dictionary (Table S5).

Positioning. The click's assignment of the second outcome from the joint weights is a non-local step on the labels, its interval the last arm's (as after Theorem 7), forced by Bell's theorem because the marginals are exact and $S > 2$ [50, 51]; no-signalling of the counts is Theorem 7 [52], and the order of the outcomes signals under the sequential wheel (Section 4.5; Table S1, row 1c, FAIL after a detector). The click's finite kinematics is Weyl's and Schwinger's on $\mathbb{Z}_{N_\phi}$ [73, 74]; the transport a lattice gas's exact integer dynamics [53, 54], free of dispersion where the lattice automata of [78, 79, 80] disperse; the uniform u does the work of quantum equilibrium [81, 82]; the model is ψ -ontic [83, 84]; Theorem 8 puts $S(N_\phi)$ on both sides of Tsirelson's bound [86, 87]. The earlier testbed [1] is history.

Declarations

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Competing interests. The author declares no competing interests.

Ethics approval. Not applicable.

Data and code availability. The simulator, every world file, every registered run's record, the check scripts with their outputs, the figures' scripts and the derivation records cited in this paper are archived at Zenodo [92] (concept DOI 10.5281/zenodo.22738746); the commit of the archived version is named in the Supplementary Material, S3, and every number of the paper has one row in the ledger of that archive (`paper/general formula/NUMBERS.md`) naming its kind and its source.

Author contributions. Sole author: the model, its rules, the worlds and the experiments, every reading and its verdict, and the decisions recorded in the archived log.

Use of artificial intelligence. A large language model (Claude, Anthropic, through Claude Code, 2026) was used as a tool under the author’s direction for code, computation and the drafting of the text; the author reviewed every statement and is responsible for every claim. The author’s statement: I laid down every postulate and hypothesis of the model myself and brought in nothing external; the forms of Newton, Kepler, Einstein, Lorentz, Bohr and Balmer that this paper cites are the things compared with, not inputs to the model.

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A Technical details

The tables and their rounding. interval $t \in \mathbb{Z}$. The circle is $\mathbb{Z}_{N_\phi}$ with $4 \mid N_\phi$. The tables are $C_{N_\phi}[p] = \text{round}(256 \cos(2\pi p/N_\phi))$ and $S_{N_\phi}[p] = C_{N_\phi}[p - N_\phi/4]$, so $C_{N_\phi}[p + N_\phi/2] = -C_{N_\phi}[p]$ and $S_{N_\phi}[p + N_\phi/4] = C_{N_\phi}[p]$ exactly; the norm $C_{N_\phi}^2 + S_{N_\phi}^2$ lies in $[65536 - 351, 65536 + 361]$ for every power of two $N_\phi \leq 65536$ [5]. The normalisation by C_K in Eq. (4) makes the click's probabilities sum to one at completion, and the paper claims exact unitarity nowhere.

The symbols. The symbol table, every letter with its kind and its name, is Table S3 of the Supplementary Material.

B Reproduction

The code, the worlds with their expectation files written before the runs, the check scripts with their outputs, the runs' summary and the figure scripts are archived [92]; the register carries every run's world, duration and source fingerprint, and NUMBERS.md beside the manuscript maps each number of this paper to its source [5]; the criterion of every row of Table S1, what is measured, its kind, what would refute it and the re-run at head, is [45]. The register's consolidation (its head `b12dd253` on the branch `register-paper-sources`, pending merge; the merge commit replaces it) keeps every world this paper cites, and the worlds it removes are cited by no row of this paper; NUMBERS.md marks the rows whose worlds the paper no longer cites. The interpreter is Python 3.14, no floating point in any physical module, the runs seconds each. To run a registered world from the archive: `python -m pip install -e .` and then `python -m event_universe --init examples/events/amplitude/bell_0_24.json --output artifacts/bell_0_24`; the worlds of this paper are under `examples/events/amplitude/`, the pair `bell_*.json`, the two slits `slits_huygens.json`, the Mach-Zehnder `mz_equal.json`, `mz_half.json` and `mz_quarter.json`; a run writes its metadata and its events beside the output, and the check scripts read them. The runs' fingerprints: the register's `ff5c382d672f` (series L), L7 `4bf55a62e6fd`, S `24e0c1ba`, T `afb533a`. The figures' runs carry the fingerprint `731d0f56c9f9` (`figures/summary.json` [5]). The archived version cited at submission is the tag `paper-2026-09-22` on the merge commit of the pull request that carries this cut, one tree whose history holds the trees named above and the log records this paper cites; it replaces the concept DOI below. The paper's law is the runtime before the generic entry of the bending: the runs' own commits named above, on the first-parent line of `main` up to `3a9a7109`, the last commit before the entry's merge `5b855791`; the generic entry is a later rule of the law (Section 5.1, P9), and the ring's run (Table S1, row 13) is this paper's one reading under it. Every record, note, design and register row this text cites is on the tree of this cut (main merged into it at `87c7ec6b`).

The long form of this manuscript, with the derivation record of every formula, is the tree's paper at the commit `c15c1174`; the forty-page cut before it at `b0d1ebf7`.

The seven confirmations. Each a run's clicks against the formula shown Inside, listed with its numbers in S3 of the Supplementary Material.